WannierBSE: User Guide

Version 1.1.0

Contents

1. Introduction

1.1 Getting Help

The latest release of ***WannierBSE*** and the associated user manual can be found at the official website: https://quantum.web.nycu.edu.tw/wannierbse/

Alternatively, the manual and the development version of the source code may be cloned or downloaded from the official repository on GitHub: https://github.com/WannierBSE

For technical issues, bug reports, or questions regarding the functionality of ***WannierBSE***, users are encouraged to utilize the official GitHub repository. You may raise specific issues or start discussions directly on the platform.

1.2 Citation

To support the continued development and maintenance of ***WannierBSE***, we kindly ask that you cite the following references in any publication resulting from the use of this software.

Primary Reference

For the general use of the ***WannierBSE*** code, please cite:

[Ref] Guan-Hao Peng, Ping-Yuan Lo, Wei-Hua Li, Yan-Chen Huang, Yan-Hong Chen, Chi-Hsuan Lee, Chih-Kai Yang, and Shun-Jen Cheng, *"Distinctive Signatures of the Spin- and Momentum-Forbidden Dark Exciton States in the Photoluminescence of Strained WSe2 Monolayers under Thermalization",* **Nano Lett.** 19, 4, 2299–2312 (2019). https://doi.org/10.1021/acs.nanolett.8b04786

BibTeX entry:

```
@article{Peng2019,
  title = {Distinctive Signatures of the Spin- and Momentum-Forbidden
Dark Exciton States in the Photoluminescence of Strained WSe2
Monolayers under Thermalization},
  author = {Peng, Guan-Hao and Lo, Ping-Yuan and Li, Wei-Hua and
Huang, Yan-Chen and Chen, Yan-Hong and Lee, Chi-Hsuan and Yang, Chih-
Kai and Cheng, Shun-Jen},
  journal = {Nano Lett.},
  volume = {19},
  issue = {4},
  pages = {2299-2312},
  year = {2019},
  doi = {10.1021/acs.nanolett.8b04786}
}
```

Non-local Dielectric Function

If you utilize the semiclassical Poisson solver to calculate the non-local dielectric function, please also cite:

[Ref] Wei-Hua Li, Jhen-Dong Lin, Ping-Yuan Lo, Guan-Hao Peng, Ching-Yu Hei, Shao-Yu Chen, and Shun-Jen Cheng, "*The Key Role of Non-Local Screening in the Environment-Insensitive Exciton Fine Structures of Transition-Metal Dichalcogenide Monolayers*", **Nanomaterials** 13, 11, 1739 (2023). https://doi.org/10.3390/nano13111739

BibTeX entry:

```
@article{Li2023,
  title = {The Key Role of Non-Local Screening in the Environment-Insensitive Exciton Fine Structures of Transition-Metal Dichalcogenide Monolayers},
  author = {Li, Wei-Hua and Lin, Jhen-Dong and Lo, Ping-Yuan and Peng, Guan-Hao and Hei, Ching-Yu and Chen, Shao-Yu and Cheng, Shun-Jen},
  journal = {Nanomaterials},
  volume = {13},
  issue = {11},
  pages = {1739},
  year = {2023},
  doi = {10.3390/nano13111739}
}
```

1.3 Credits

The ***WannierBSE*** software package was developed by the research group of Prof. Shun-Jen Cheng at National Yang Ming Chiao Tung University (NYCU). The project is supervised and guided by Prof. Shun-Jen Cheng.

Core Developers:

- Dr. Ping-Yuan Lo
- Dr. Wei-Hua Li
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- Dr. Jhen-Dong Lin
- Dr. Vo Khuong Dien
- Dr. Oscar Javier Gomez Sanchez
- Mr. Ching-Hung Shih
- Mr. Kun-Yi Lin
- Prof. Shun-Jen Cheng

1.4 License

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2. Theoretical Framework

WannierBSE is an efficient computational package for solving the Bethe-Salpeter equation (BSE) for neutral exciton in 2D materials within the Wannier tight-binding (WTB) framework based on the density-functional theory (DFT), implemented in the MATLAB language. The theoretical foundation of the BSE within this framework is based on the many-body formalisms developed by the seminal works of Refs. [1-7]. Technically, it takes advantage of global adaptive integration algorithms for fast and accurate construction of electron-hole (e-h) Coulomb kernels of BSE.

The core of the package relies on a multi-scaled methodology that integrates a microscopic Hamiltonian based on Wannier functions [8,9] with a macroscopic Poisson solver to calculate the screened Coulomb interaction.

WannierBSE can efficiently construct the BSE for neutral exciton in 2D materials based on the DFT-calculated band structures and Bloch wavefunctions, and dielectric screening from the DFT+GW or macroscopic electromagnetic theory. This flexibility enables ***WannierBSE*** to perform multiscale simulations of exciton spectra in 2D materials, capturing the effects of realistically complex dielectric environments.

The ***WannierBSE*** package is developed to numerically solve the Bethe-Salpeter equation (BSE) [3,10,11],

$$\begin{aligned}(\epsilon_{c\boldsymbol{k}+\boldsymbol{k}_{ex}} - \epsilon_{v\boldsymbol{k}})A_{S\boldsymbol{k}_{ex}}(vc\boldsymbol{k}) &- \sum_{v'c'\boldsymbol{k}'}\big[V^{eh,d}_{\boldsymbol{k}_{ex}}(vc\boldsymbol{k}, v'c'\boldsymbol{k}') \\ &- V^{eh,x}_{\boldsymbol{k}_{ex}}(vc\boldsymbol{k}, v'c'\boldsymbol{k}')\big]A_{S\boldsymbol{k}_{ex}}(v'c'\boldsymbol{k}') = E_{S\boldsymbol{k}_{ex}}A_{S\boldsymbol{k}_{ex}}(vc\boldsymbol{k}),\end{aligned} \tag{2.1}$$

where $\epsilon_{\mathrm{n}k}$ is the kinetic energy, $\mathrm{E}_{S\boldsymbol{k}_{ex}}$ is the energy of exciton state labeled by quantum number S with center-of-mass (CoM) wave vector $\boldsymbol{k}_{ex}$**.** The e-h direct and exchange Coulomb kernels read [5,6,12,13],

$$V^{eh,d}_{\boldsymbol{k}_{ex}}(vc\boldsymbol{k}, v'c'\boldsymbol{k}') = \int\int d^3\boldsymbol{r}_1 d^3\boldsymbol{r}_2\, \psi^*_{c\boldsymbol{k}+\boldsymbol{k}_{ex}}(\boldsymbol{r}_1)\psi^*_{v'\boldsymbol{k}'}(\boldsymbol{r}_2)W(\boldsymbol{r}_1,\boldsymbol{r}_2)\psi_{v\boldsymbol{k}}(\boldsymbol{r}_2)\psi_{c'\boldsymbol{k}'+\boldsymbol{k}_{ex}}(\boldsymbol{r}_1), \tag{2.2}$$

$$V^{eh,x}_{\boldsymbol{k}_{ex}}(vc\boldsymbol{k}, v'c'\boldsymbol{k}') = \int\int d^3\boldsymbol{r}_1 d^3\boldsymbol{r}_2\, \psi^*_{c\boldsymbol{k}+\boldsymbol{k}_{ex}}(\boldsymbol{r}_1)\psi^*_{v'\boldsymbol{k}'}(\boldsymbol{r}_2)V(\boldsymbol{r}_1 - \boldsymbol{r}_2)\, \psi_{c'\boldsymbol{k}'+\boldsymbol{k}_{ex}}(\boldsymbol{r}_2)\psi_{v\boldsymbol{k}}(\boldsymbol{r}_1). \tag{2.3}$$

Here, $\psi_{\mathrm{n}\boldsymbol{k}}(r)$ is the single-electron wavefunction, $W(\boldsymbol{r}_1,\boldsymbol{r}_2) = \int \mathrm{d}^3\mathbf{r}'\varepsilon^{-1}(\mathbf{r}_1,\mathbf{r}')V(\boldsymbol{r}' - \boldsymbol{r}_2)$ is the screened Coulomb potential, with $\varepsilon^{-1}(\boldsymbol{r}_1,\boldsymbol{r}')$ being the inverse non-local dielectric function and $V(\mathbf{r}) = \frac{e^2}{4\pi\varepsilon_0|\mathbf{r}|}$ being the bare Coulomb potential.

By formulating the e-h Coulomb kernels in the WTB framework, which unitarily connects the single-electron Bloch wavefunctions with the maximally localized Wannier function (MLWF) basis $\mathcal{W}_j(\boldsymbol{\rho}-\boldsymbol{R},z) = \langle\boldsymbol{\rho}-\boldsymbol{R},z|\mathcal{W}_{j\boldsymbol{R}}\rangle$ by [14]

$$\psi_{n\boldsymbol{k}}(\boldsymbol{r}) = \frac{1}{\sqrt{N}}\sum_j\sum_{\boldsymbol{R}} C_j^{(n)}(\boldsymbol{k})e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\mathcal{W}_j(\boldsymbol{\rho}-\boldsymbol{R},z), \tag{2.4}$$

where j labels the orbital-like WFs, $\boldsymbol{R}$ indicates the unit cell that the WF locates at, and the coefficients $C_j^{(n)}(\boldsymbol{k})$ are determined by the WTB Hamiltonian

$$H_{ij}^{WTB}(\boldsymbol{k}) = \sum_{\boldsymbol{R}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}}\langle\mathcal{W}_{i\boldsymbol{0}}|\hat{H}^{KS}|\mathcal{W}_{j\boldsymbol{R}}\rangle, \tag{2.5}$$

which is equivalent to the Kohn-Sham Hamiltonian $\hat{H}^{KS}$ represented in the WF basis.

Within the WTB framework, the Bloch wavefunctions $\psi_{nk}(r)$ can be expanded by a small set (typically 10^1~10^2) of MLWFs, significantly reduces the computational costs of constructing e-h Coulomb kernels of the BSE. Thus, by using the highly localized Wannier functions, the electron-hole direct interaction is re-formulated as,

$$\begin{aligned} &V_{\boldsymbol{k}_{ex}}^{eh,d}(vc\boldsymbol{k},v'c'\boldsymbol{k}') \\ &\quad= \frac{1}{\mathcal{A}}\sum_j\sum_\ell C_j^{(c')}(\boldsymbol{k}'+\boldsymbol{k}_{ex})C_j^{(c)*}(\boldsymbol{k}+\boldsymbol{k}_{ex})C_\ell^{(v')*}(\boldsymbol{k}')C_\ell^{(v)}(\boldsymbol{k}) \\ &\qquad\qquad \times e^{i(\boldsymbol{k}-\boldsymbol{k}')\cdot(\tau_j-\tau_\ell)}W(\boldsymbol{k}-\boldsymbol{k}'), \end{aligned} \tag{2.6}$$

Here, τ_j denotes the atomic position vector where the Wannier function is localized. The term $W(q) = V(q)/\epsilon(q)$ corresponds to the screened Coulomb interaction in the momentum space with $V(q)$ the unscreened Coulomb interaction [19].

On the other side, the electron-hole exchange interaction is re-formulated as

$$\begin{aligned} &V_{\boldsymbol{k}_{ex}}^{eh,x}(vc\boldsymbol{k},v'c'\boldsymbol{k}') \\ &\quad= \sum_{jj'}\sum_{\ell\ell'}\sum_{RR'} C_j^{(v)}(\boldsymbol{k})C_{j'}^{(c)*}(\boldsymbol{k}+\boldsymbol{k}_{ex})C_{\ell'}^{(v')*}(\boldsymbol{k}')C_\ell^{(c)}(\boldsymbol{k}'+\boldsymbol{k}_{ex}) \\ &\qquad\qquad \times e^{-i(\boldsymbol{k}+\boldsymbol{k}_{ex})\cdot\boldsymbol{R}}e^{i(\boldsymbol{k}'+\boldsymbol{k}_{ex})\cdot\boldsymbol{R}'}\chi_{j'j\ell'\ell}^{\boldsymbol{RR}'}, \end{aligned} \tag{2.7}$$

Where $\chi_{j'j\ell'\ell}^{\boldsymbol{RR}'}$ is given by

$$\chi_{j'j\ell'\ell}^{\boldsymbol{RR}'} = \frac{e^2}{2\epsilon_0}\sum_G \frac{1}{|\boldsymbol{k}_{ex}+\boldsymbol{G}|}\Xi_{j'j\ell'\ell}^{\boldsymbol{RR}'}(\boldsymbol{k}_{ex}+\boldsymbol{G}), \tag{2.8}$$

with

$$\begin{aligned} &\Xi_{j'j\ell'\ell}^{\boldsymbol{RR}'}(\boldsymbol{k}_{ex}+\boldsymbol{G}) \\ &\quad\equiv \int dz_1dz_2\varrho_{j'j}^{(\boldsymbol{R})*}(\boldsymbol{k}_{ex}+\boldsymbol{G};z_1)\varrho_{\ell'\ell}^{(\boldsymbol{R}')}(\boldsymbol{k}_{ex}+\boldsymbol{G};z_2)\,e^{-|\boldsymbol{k}_{ex}+\boldsymbol{G}||z_1-z_2|}, \end{aligned} \tag{2.9}$$

Here, $\boldsymbol{G}$ is the reciprocal lattice vector and $\Xi^{RR'}_{j'jl'l}(k_{ex}+G)$ defined as

$$\varrho^{(\boldsymbol{R})}_{j'j}(\boldsymbol{k}_{ex}+\boldsymbol{G};z) \equiv \int d\boldsymbol{\rho}\, e^{-i(\boldsymbol{k}_{ex}+\boldsymbol{G})\cdot\boldsymbol{\rho}} \mathcal{W}^{*}_{j'}(\boldsymbol{\rho},z)\mathcal{W}_{j}(\boldsymbol{\rho}-\boldsymbol{R},z). \tag{2.10}$$

For further technical details, see Peng et al., *Nano Lett.* 2019, 19, 2299–2312 [14].

Note on Version 1.1: *Based on the electron-hole (e-h) separation in real space, the exchange Coulomb interaction shown in equation (2.7) can be divided into short- and long-range components. In transition metal dichalcogenide monolayers (TMD-MLs), the short-range exchange increases the energy of spin-allowed excitons [3, 15], while the long-range exchange drives the mixing of intervalley excitons, leading to an energy splitting of the spin-allowed states [3,16]. In the current release of* ***WannierBSE*** *(v1.1), the long-range electron-hole exchange interaction is not considered, as the package focuses primarily on the direct interaction term, which dominates the exciton binding energy [6,13], alongside the short-range electron-hole exchange interaction, which in momentum space corresponds to* $G\neq 0$. *Furthermore, the implementation strictly considers excitons with zero center-of-mass momentum,* $k_{ex}=0$, *though future versions of the package are planned to incorporate the long-range component of the exchange kernel and excitons with finite momentum.*

2.1 Dielectric Screening

The accurate description of excitonic properties in TMD-MLs is fundamentally linked to the treatment of dielectric screening. The methodology employed here for modeling non-local screening and the dielectric environment of 2D semiconductors builds upon the influential work of Ref. [5,6,13,17,18]. Due to reduced dimensionality, TMDs exhibit weak intrinsic screening, leading to enhanced e-h interactions. In ***WannierBSE***, the dielectric environment is treated by solving the Poisson equation for a layered structure, accommodating encapsulated or substrate-supported monolayers. This approach allows for the accurate description of excitonic complexes embedded in multi-layered dielectric environments, accounting for the non-local screening effects arising from the surrounding media.

The screened Coulomb interaction $W(r_1,r_2)$ is essentially the Green's function describing the potential at r_1 due to a point charge at r_2. In the presence of a non-local dielectric function $\varepsilon^{-1}(\boldsymbol{r}_1,\boldsymbol{r}')$, the interaction obeys the general form of the Poisson equation [6,13,18-20]:

$$\boldsymbol{\nabla}_{\boldsymbol{r}_1}\cdot\int \mathrm{d}^3\mathbf{r}'\varepsilon^{-1}(\mathbf{r}_1,\mathbf{r}')\boldsymbol{\nabla}_{\boldsymbol{r}'}W(\boldsymbol{r}',\boldsymbol{r}_2) = -\frac{\mathrm{e}^2}{\varepsilon_0}\delta(\boldsymbol{r}_1-\boldsymbol{r}_2). \tag{2.11}$$

To exploit the lattice translational symmetry of the 2D plane, the equation is reformulated using an in-plane Fourier expansion. Under a long-range approximation, the system is modeled as a vertical heterostructure where the dielectric environment and the TMD layer are treated as distinct regions with specific boundary conditions.

WannierBSE employs a piece-wise dielectric function to model the vertical heterostructure. A typical configuration consists of a TMD monolayer of effective thickness d embedded in a dielectric environment (e.g., air, hBN, or SiO_2). Within the monolayer region ($-d/2 < z < d/2$), the screening is characterized by a q-dependent bulk dielectric function, which in the random-phase approximation is given by [17,18]:

$$\varepsilon_{\mathrm{TMD}}(q) = 1 + \frac{1}{(\varepsilon_{2D}-1)^{-1}+\alpha\frac{q^2}{q_{TF}^2}+\left(\frac{\hbar^2 q^2}{2m_0 E_{pl}}\right)^2}. \tag{2.12}$$

In this expression, ε_{2D} is the in-plane dielectric constant of the bulk TMD, α is a dimensionless fitting parameter, q is the in-plane momentum, and E_{pl} represents the plasma peak energy. The non-locality of the screening is further governed by the Thomas-Fermi screening wave vector, q_{TF}, which is mathematically defined as [13,18]:

$$q_{TF} = \sqrt{\frac{e^2 m_0}{\pi^2 \epsilon_0 \hbar^2}\left(\frac{3\pi^2 \varepsilon_0 m_0 E_{pl}^2}{e^2\hbar^2}\right)^{\frac{1}{3}}}. \tag{2.13}$$

The resulting effective dielectric function of the system is defined as [5,6]

$$\varepsilon(\boldsymbol{q}) = \frac{\overline{V}(\boldsymbol{q})}{\overline{W}(\boldsymbol{q})}. \tag{2.14}$$

Here, $\overline{V}(\boldsymbol{q})$ represents the averaged unscreened interaction, and $\overline{W}(\boldsymbol{q})$ is the effective 2D screened potential averaged over the thickness d. The coefficients required to determine the final potential are calculated by matching boundary conditions at the interfaces, ensuring the continuity of the potential and the displacement field across the heterostructure boundaries. For a complete derivation of these expressions and the specific parameters for various TMD materials, see Li et al., *Nanomaterials* **2023**, *13*, 1739 [21].

3. WannierBSE: Software Implementation

WannierBSE v1.1 is an advanced computational suite designed to solve the Bethe-Salpeter Equation (BSE) for neutral excitons in two-dimensional materials. By bridging microscopic Wannier-based tight-binding models with a macroscopic treatment of non-local dielectric screening, the software provides a robust framework for investigating excitonic properties. To accommodate various research objectives, the software offers two primary operational workflows governed by the chosen Hamiltonian interactions:

- Direct-Interaction Mode: This mode focuses exclusively on the screened direct electron-hole Coulomb interaction. It offers maximum structural flexibility, allowing users to either employ the automated, internal Wannier90-based workflow where the user provides only the fundamental Wannier Hamiltonian (`wannier90_hr.dat`), and the package automatically resolves the required momentum grid and band structure; or Users may bypass internal solvers by providing their own pre-calculated external Tight-Binding models.

- Short-Range Exchange Mode: This mode incorporates both the screened direct interaction and the short-range electron-hole exchange interaction. Because the exchange kernel requires a consistent spinor Wannier basis, real-space Wannier functions, and internally generated exchange tensors, this mode is strictly restricted to the internal Wannier90-based workflow. Consequently, enabling short-range exchange additionally requires the real-space spinor Wannier functions from XSF files to compute the necessary exchange tensors.

The resolution of dielectric screening remains entirely independent of the chosen electronic structure path. In either operational mode (direct-only or with short-range exchange), the dielectric screening data can be resolved via two routes: the user may manually provide a custom dielectric function via `User_input/epsilon.txt`, or allow ***WannierBSE*** to automatically invoke its internal macroscopic Poisson solver to generate `Precomputed_data/epsilon_WBSE.txt` based on the parameters specified in `Parameters/dielectric_control.txt`.

To manage these workflows effectively, ***WannierBSE*** utilizes a modular directory architecture that strictly separates user configurations, raw inputs, internal operations, and final assets into five main areas:

- **Parameters**: Serves as the central repository for the configuration files that dictate the physical and numerical execution parameters of the simulation. It contains the core runtime controls (control.txt), tight-binding parameters (`WTB_control.txt`), k-mesh configurations (`kmesh_control.txt`), and screening variables (`dielectric_control.txt`), alongside the newly implemented short-range exchange parameters (`exchange_control.txt`) and the three-column, spinor-resolved Wannier center configurations (`WF_centers.txt`).

- **User_input**: Actively functions as the primary gateway for all user-supplied physical datasets and structural definitions. It ingests lattice and atomic geometries (preferring `wannier90.win` or a simplified `structure.txt` file), standard Wannier90 tight-binding matrices (`wannier90_hr.dat`), and optional standalone electronic data (`kmesh.txt`, `epsilon.txt`, or paired valence/conduction matrices within `TB_data/`). For short-range exchange calculations requiring real-space evaluation, it also hosts raw spinor-resolved spatial wavefunctions organized within the dedicated `Wannier_functions_xsf/up_real/`, `Wannier_functions_xsf/up_imag/`, `Wannier_functions_xsf/down_real/`, and `Wannier_functions_xsf/down_imag/` subdirectories.

- **Precomputed_data**: Functions exclusively as ***WannierBSE***'s internal cache and must not be utilized for user-defined inputs. The software systematically writes its internally generated, intermediate physics representations here, such as synchronized k-meshes (`kmesh_WBSE.txt`), dielectric matrices (`epsilon_WBSE.txt`), and spin-block ordered bases (`WF_centers_WBSE.txt`), the dedicated `TB_data/` subfolder containing cached valence/conduction band structures, alongside heavy-duty binary assets like processed real-space spinor matrices under `Wannier_functions_WBSE/` (`WF_up.mat`, `WF_down.mat`) and pre-calculated exchange interaction tensors under `Exchange_interaction/` (`chi_WBSE.mat`, `R_mNN_WBSE.mat`).

- **Exciton_data**: Centralizes the final, publication-ready physical observables and complete execution records generated at the end of a simulation run. This folder archives the solved excitonic eigenvalues (`Ex.mat`), eigenvectors (`A.mat`), momentum grids (`k.mat`), and execution benchmarks/log files such as `wi.mat` and `WBSE_log.txt`. Furthermore, depending on the active workflow, it dynamically encapsulates tracked traceability assets, such as user-style rearranged tight-binding records stored within `TB_Bands/` or intermediate exchange overlap densities and tensors housed inside `Exchange_data/` (`rho.mat`, `chi.mat`).

- **Functions_for_WBSE**: Houses the entire monolithic computational infrastructure, core kernels, and specialized loaders required to execute the Bethe-Salpeter calculation. Its internal architecture is split into functional subfolders that govern data ingestion and parameter parsing (`Loaders/`, `Utils/`), initial matrix setup (`TB/`, `Kmesh/`, `Dielectric/`), heavy preprocessing engines (`Wannier_engine/`), and the final physics evaluation routines (`Core/`, containing `Coulomb/`, `Direct/`, and `Exchange/` sub-kernels) which handle the formal numerical assembly of the final BSE Hamiltonian matrix.

Upon execution, ***WannierBSE*** processes physical data and parameters using a structured priority logic where explicit user inputs take precedence over internal caches, which in turn override automated generation engines (User Input > Internal Cache > Automated Generation)

For the macroscopic dielectric function, the software first checks for a user-provided `User_input/epsilon.txt`. If absent, it looks for cached data in the

Precomputed_data directory, and failing that, it automatically invokes its internal Poisson solver to generate the screening data based on the parameters defined in `dielectric_control.txt`. This dielectric screening workflow remains completely independent of how the electronic structure is resolved.

For the electronic structure, k-meshes and tight-binding bands follow a similar prioritization but are strictly governed by rigorous consistency constraints. If a complete, matched pair of `User_input/kmesh.txt` and a full set of valence and conduction band files in `User_input/TB_data/` is provided, the software ingests them directly and will ignore raw Wannier90 files present in the input directory. However, providing only one of these external elements or an incomplete set of band files causes the software to immediately terminate with an input-consistency error, as mixing partial external data with internal generation risks silent k-point and coefficient mismatches. Furthermore, the internal k-mesh and tight-binding band cache stored in the `Precomputed_data/` directory are also treated as a single coherent group; if the software detects that this internal cache is incomplete or stale relative to the governing control files, it will systematically rebuild the entire k-mesh and tight-binding cache to maintain perfect synchronization.

Crucially, the use of custom, user-provided external electronic models is strictly restricted to direct-interaction-only calculations and is completely blocked when the short-range electron-hole exchange interaction is enabled. Attempting to supply external k-mesh and tight-binding data while *Exchange_Interaction = true* will trigger an error, because the exchange kernel fundamentally requires a consistent spinor Wannier basis, real-space Wannier functions and internally generated exchange tensors that must derive exclusively from the internal Wannier90 workflow. Consequently, when exchange is active, the software bypasses external electronic inputs entirely, relying instead on the internal path to preprocess the necessary real-space spinor data and consistently generate or load the coupled tight-binding and exchange interactions required to build the full Bethe-Salpeter Hamiltonian.

3.1 Parameters

The execution and physical behavior of ***WannierBSE*** v1.1 are governed by a set of plain-text configuration files located within the `Parameters/` directory. These files define the complete numerical and physical environment of the Bethe-Salpeter Equation (BSE) solver, including active band windows, momentum-space grid generation rules, dielectric screening profiles, Wannier-function basis conventions, and short-range electron-hole exchange interactions.

In ***WannierBSE*** v1.1, the direct electron-hole Coulomb interaction forms the fundamental baseline of the BSE Hamiltonian and is always active. Conversely, the short-range exchange interaction is a modular feature that is activated and configured explicitly via the `exchange_control.txt` file.

The standard parameter files driving the computational pipeline comprise:

- **control.txt**: The primary configuration file regulating global simulation settings. It specifies the effective material thickness, the number of target exciton eigenvalues to be solved, the active valence and conduction band windows, and parallelization parameters tailored to optimize the various computational stages.

- **WF_centers.txt**: Establishes the spatial mapping between Wannier functions and their respective atomic centers. In version 1.1, this file is expanded to define the spinor component of each Wannier function, a mandatory requirement for executing the Wannier90-based spinor workflow and performing short-range exchange calculations.
- **exchange_control.txt**: Manages the inclusion of the short-range electron-hole exchange interaction. This file defines critical algorithmic variables, including exchange cutoff parameters, the real-space Wannier-function truncation threshold, and configuration toggles to reuse precomputed exchange tensors for accelerated calculation restarts.
- **WTB_control.txt**: Required when ***WannierBSE*** operates in its automated workflow to construct tight-binding band structures directly from the `wannier90_hr.dat` file. It designates the valence-band maximum (VBM) and conduction-band minimum (CBM) indices necessary for isolating the active energy window.
- **kmesh_control.txt**: Triggered conditionally when a user-defined momentum grid (`User_input/kmesh.txt`) is absent. It dictates the internal generation routines for constructing a non-uniform k-point mesh across the Brillouin zone.
- **dielectric_control.txt**: Required conditionally when an external screening profile (`User_input/epsilon.txt`) is not provided. It prescribes the multi-layer substrate architecture and dielectric constants required by the internal macroscopic Poisson solver to generate the non-local dielectric screening function.

The exact combination of required parameter files depends strictly on the selected operational workflow. When executing a calculation in Direct-Interaction Mode, users maintain the flexibility to either utilize the fully automated internal Wannier90 generation workflow or bypass it by providing a pre-matched external dataset consisting of `kmesh.txt` and a `TB_data/` directory. However, when operating in Direct plus Exchange-Interaction Mode, the software mandates the internal Wannier90-based workflow. This requirement ensures that the exchange kernel has direct access to a mutually consistent spinor Wannier basis and the real-space Wannier localization data necessary for the calculation.

In the following paragraphs, the exact syntax, data formatting, and specific variable scopes for each control file are detailed.

3.1.1 General Control: control.txt

This file defines the fundamental computational settings for the BSE calculation, including the active band window and parallelization preferences. Please note that the MATLAB Parallel Computing Toolbox is required to run these calculations.

Parameter Description:

Keyword	Type	Description
d	Real	The effective thickness of the 2D material slab, specified in Angstroms (Å). This value is used for the truncation of the Coulomb interaction in the direction perpendicular to the layer.
NVAL	Integer	The number of lowest-energy exciton eigenstates (eigenvalues) to compute during the diagonalization of the BSE Hamiltonian.
Nc	Integer	The number of conduction bands to include in the electron-hole basis set. *(Note: This must be less than or equal to the number of* `c*_TB.txt` *input files provided).*
Nv	Integer	The number of valence bands to include in the electron-hole basis set. *(Note: This must be less than or equal to the number of* `v*_TB.txt` *input files provided).*
Npar_TB	Integer	Number of workers used for Wannier tight-binding band generation. This parameter is relevant when band data is generated internally from `wannier90_hr.dat`.
Npar_rho	Integer	Number of workers used to compute the short-range exchange overlap density rho. This parameter is used only when short-range exchange is enabled and the exchange tensors are computed during the run.
Npar_Xi	Integer	Number of workers used to evaluate the Ξ integrals that enter the short-range exchange tensor χ. This parameter is used only when short-range exchange is enabled and the exchange tensors are computed during the run.
Npar_HBSE	Integer	Number of workers used for BSE Hamiltonian construction. This is the main parallelization parameter for the BSE solver stage.

Note: *The number of allocated cores specified for any computational stage must not exceed the total physical CPU core count available on the local machine. For laptops or memory-constrained systems, setting this parameter to a minimal value (typically 1) will deactivate the parallel execution pool entirely, preventing potential memory exhaustion and conserving system hardware resources.*

Example File:

```
d      = 6.26 % 2D material thickness in Angstroms
%---------------BSE calculation options--------------------------
NVAL    = 100 % Number of exciton eigenvalues to find
Nc      = 2   % Number of conduction bands in BSE
Nv      = 2   % Number of valence bands in BSE
%---------------parallel calculation parameters------------------
Npar_TB  = 12 % Number of cores for tight-binding band generation
Npar_rho = 40 % Number of cores for exchange overlap density rho
Npar_Xi  = 40 % Number of cores for exchange Xi integrals
Npar_HBSE= 40 % Number of cores for BSE Hamiltonian construction
```

3.1.2 Wannier Function Centers: WF_centers.txt

The WF_centers.txt file maps Wannier basis functions to their corresponding atomic centers and spinor components. This mapping ensures consistency across the tight-binding basis, phase factors, and the exchange workflow.

The file is a space-separated, plain-text format structured into three sequential columns:

- Column 1: Wannier-function Index [Integer]. Represents the unique identifier for each Wannier function. The indices must be strictly contiguous and ordered sequentially from 1 up to the total number of Wannier functions included in the calculation.
- Column 2: Atomic Index [Integer]. The index of the atomic center where the Wannier function is localized. This must match the exact sequence of atoms defined in the atoms_cart block of `User_input/structure.txt` or `User_input/wannier90.win`.
- Column 3: Spinor Component [String]. Designates the spinor orientation of the localized Wannier function. Valid standard entries are explicitly restricted to lowercase labels: up or down.

To verify the correct correlation between each Wannier-function index and its corresponding atomic index, users should cross-reference the standard `wannier90.wout` output file. This file explicitly details the calculated centers of the maximized Wannier functions alongside their nearest atomic positions, ensuring that assignments match the sequential ordering established in the structural configuration files.

Recommendation**:** It is strongly advised to visually plot the calculated Wannier functions to confirm spatial localization prior to executing the BSE solver.

Example File:

```
 i      I      Spinor
 1      2      up
 2      2      down
 3      2      up
 4      2      down
 ...
21      1      up
22      1      down
```

3.1.3 Short-Range Exchange Control: exchange_control.txt

The `exchange_control.txt` file determines whether ***WannierBSE*** incorporates the short-range electron-hole exchange interaction into the core BSE Hamiltonian.

Additionally, it defines the numerical thresholds and real-space truncation cutoffs required to construct the exchange tensors from spinor Wannier functions.

This short-range exchange workflow is strictly compatible with the automated, Wannier90-based calculation pipeline. External, user-provided electronic models (such as standalone `kmesh.txt` and `TB_data/` configurations) are restricted to direct-interaction simulations and are incompatible with the exchange framework; setting *Exchange_Interaction = true* with external models will result in an execution error. Setting *Exchange_Interaction = false* or in the absence of `exchange_control.txt` within the `Parameters/` directory, the solver automatically bypasses the exchange module and defaults to a direct-interaction-only simulation.

Keyword	Type	Description
Exchange_Interaction	Boolean	Enables or disables the short-range electron-hole exchange interaction. When set to true, the BSE Hamiltonian integrates the exchange kernel alongside the screened direct Coulomb interaction, capturing critical exchange-induced energy shifts and corrections in the exciton spectrum
Use_Precomputed_Chi	Boolean	Controls whether the solver loads a pre-existing exchange tensor from the internal cache or evaluates it from scratch during the current run. Setting this parameter to true instructs ***WannierBSE*** to retrieve `chi_WBSE.mat` and `R_mNN_WBSE.mat` from the `Precomputed_data/Exchange_interaction/` directory. This significantly accelerates subsequent simulations, provided the cached files are fully consistent with the active Wannier basis and crystal structure.
cleanup_xsf	Boolean	Specifies whether intermediate XCrysden Structure Files (XSF) generated during Wannier-function preprocessing are automatically purged. Disabling this feature (false) preserves these files for spatial inspection or visualization of the real-space Wannier functions, though it increases disk space consumption.
M_max	Integer	Defines the maximum M-nearest-neighbor shell of Wannier-function center pairs to include in the short-range exchange summation. Increasing this integer encompasses more distant Wannier-function pairs, expanding the real-space range of the exchange kernel

		at the expense of higher computational overhead.
G_max	Real	Establishes the plane-wave cutoff for the reciprocal lattice vectors (G) utilized in evaluating the exchange interaction. Higher values incorporate finer spatial-frequency components of the exchange kernel, enhancing numerical accuracy for short-range features while scaling up memory allocation and computation time.
tol_probability	Real	A real-space probability threshold used to truncate the spatial localization boxes of individual Wannier functions. Values approaching 1.0 (such as 0.98) retain a larger spatial volume of each Wannier function, yielding highly accurate exchange integrals by widening the real-space integration domains.
M_max tolerance	Real	A distance-based tolerance threshold used during the identification of the m-nearest-neighbor Wannier-function pairs bounded by M_max. This parameter (parsed internally by the core engine as M_max_tolerance) absorbs minor numerical variations in atomic coordinates or Wannier centers, preventing artificial misclassifications of adjacent neighbor shells.
nn_distance_tolerance	Real	Specifies the numerical tolerance window for classifying nearest-neighbor shells among Wannier-function centers. It regulates how strictly centers are grouped into discrete shells and should be kept sufficiently small to prevent physically distinct shells from bleeding into one another.

Example File:

```
% --- Exchange Calculation Options ---

Exchange_Interaction = true % Enable short-range electron-hole
exchange interaction
Use_Precomputed_Chi = false % Load precomputed chi/R_mNN instead of
                              recalculating them
cleanup_xsf = false % Keep raw XSF files for inspection or visualization

% --- Short Range Exchange Parameters ---
```

```
M_max = 1   % Maximum m-nearest-neighbor shell for exchange pairs
G_max = 4.2 % Reciprocal-space G-vector cutoff for exchange interaction
tol_probability = 0.98 % Probability cutoff for truncating Wannier-
                         function boxes
M_max tolerance = 0.01 % Distance tolerance for identifying m-NN exchange
                         pairs
nn_distance_tolerance = 0.15 % Tolerance for nearest-neighbor distance
                               classification
```

3.1.4 K-Mesh Generation: kmesh_control.txt

This configuration file defines the parameters for generating the momentum space grid (k-mesh) used in the BSE calculation. The ***WannierBSE*** code invokes the mesh generation routine only if a pre-existing `kmesh.txt` file is not found in the `User_input/` directory. If `kmesh.txt` is already present, this control file is ignored.

The generation algorithm constructs a non-uniform grid characterized by a dense region and a coarse region, see Figure 1. The base resolution of the global Brillouin zone is defined by Nk_coarse, which sets the geometric split along the line from the Γ point to the K point. This parameter effectively determines the mesh fineness for the bulk of the Brillouin.

The parameter Dense_range acts as an integer factor to determine the coverage of the dense region in units of the coarse grid. It specifies the spatial extent of high-resolution sampling centered at the high-symmetry K and K' points. To maintain geometric consistency and avoid overlapping points between adjacent valleys, this value must satisfy the constraint, $Dense\_range \leq Nk\_coarse/2$. The internal grid density within these refined regions is governed by Nk_dense, which represents the number of k-points sampled along a single direction within the dense sub-mesh.

Example File:

```
% ---  Control parameters for k-mesh generation  ---

Nk_coarse= 6        % Number of k-points along one direction in
the coarse region.

Nk_dense = 4        % Number of k-points along one direction in
the dense region.

Dense_range = 2     % Integer factor to determine dense region
                      size in units of the coarse grid.
```

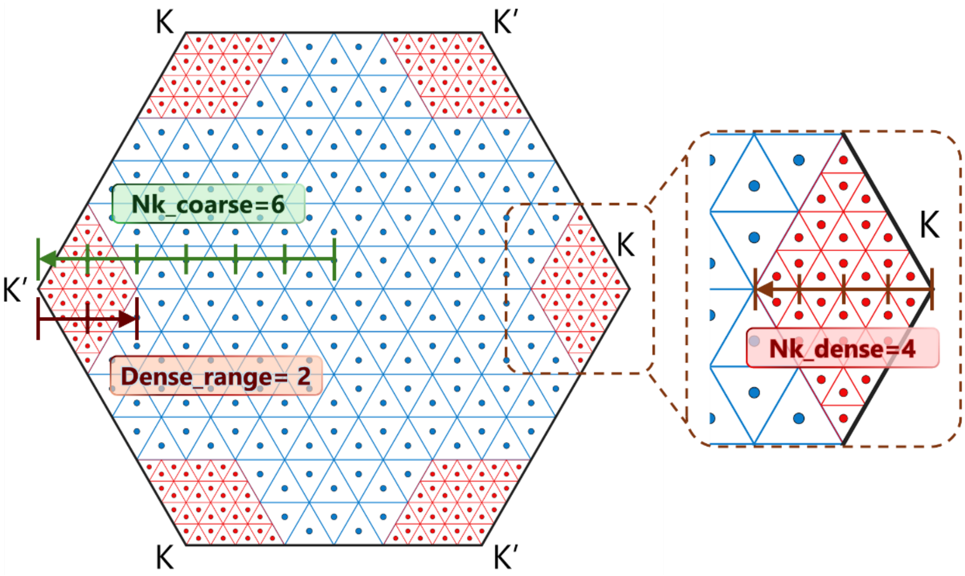


Figure 1. Schematic of the non-uniform k-mesh. The Brillouin zone is partitioned into a coarse grid (blue) for the global domain and a high-density grid (red) localized around the K and K' valleys. The overall coarse resolution is determined by Nk_coarse = 6, which sets the geometric split along the line from the Γ point to the K point. The parameter Dense_range = 2 acts as an integer factor to determine the coverage of the dense region in units of the coarse grid, as illustrated, it allocates two-sixths (2/6) of the total the Γ to the K distance to the high-resolution zones. The internal sampling density within these refined regions is controlled by Nk_dense = 4, allowing for enhanced precision at high-symmetry points while maintaining computational efficiency throughout the remaining Brillouin zone.

3.1.5 Wannier Tight-Binding Control: WTB_control.txt

This file is required when the system Hamiltonian is constructed using the `wannier90_hr.dat` hopping parameter file generated by Wannier90. Since the Wannier90 output often contains a large number of bands, it is necessary to explicitly identify the indices that define the semiconductor band gap.

Optical transitions are most likely to occur between the bands in the immediate vicinity of the band gap. Therefore, the parameters in this file serve as reference points for the band selection process. The parameter VBM (Valence Band Maximum) identifies the index of the top-most valence band, while CBM (Conduction Band Minimum) identifies the index of the lowest conduction band.

Example File:

```
% Control parameters for Wannier Tight-Binding Calculation

VBM = 14              % VBM: Valence Band Maximum index
CBM = 15              % CBM: Conduction Band Minimum index
```

3.1.6 Dielectric Model Control: dielectric_control.txt

In the absence of a pre-calculated `epsilon.txt` file, ***WannierBSE*** utilizes an internal Poisson solver to generate the necessary dielectric function. This file provides the geometric and material parameters required to construct the dielectric environment model.

The model assumes a multi-layer heterostructure. The code constructs the stack based on the number of layers defined above (t) and below (s) the central TMD monolayer, see Figure 2.

- Total Layers: The total number of dielectric regions, including the TMD monolayers, is given by $N_{total} = t + 1 + s$.
- Boundaries: There must be $s + t$ interfaces (boundaries) defined to separate these regions.
- Ordering: All lists (dielectric constants and boundaries) must be specified from Top to Bottom (positive z to negative z).

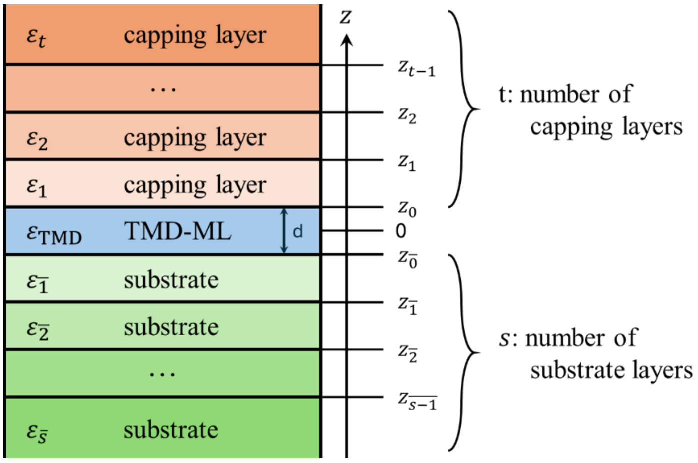


Figure 2: Schematics of the multilayer structure with TMD embedded in the middle. The TMD-ML of thickness d is embedded within a stack of capping t and substrate s layers, resulting in a spatially dependent, piecewise dielectric function. The electronic environment is characterized by the q-dependent bulk dielectric function of the monolayer, $\varepsilon_{TMD}(q)$, while the surrounding media are defined by their respective dielectric constants, ε_t for the capping layers and $\varepsilon_{\bar{s}}$ for the substrate layers. The vertical z-coordinates (z_n) denote the discrete interfaces between adjacent dielectric regions.

Parameter Description:

Keyword	Type	Description
alpha	Real	A dimensionless fitting parameter associated with the Rytova-Keldysh potential.
Epl	Real	The plasmon peak energy, specified in eV.
ep_p	Real	The specific in-plane dielectric constant of the TMD monolayer.

t_layers	Integer	The number of dielectric layers situated above the TMD monolayer.
s_layers	Integer	The number of dielectric layers situated below the TMD monolayer.
t_dielectric	List	A comma-separated list of dielectric constants for the layers above the TMD monolayer. Following the convention in Figure 2, the list is ordered starting from the top layer: t_dielectric= $\varepsilon_t, \cdots, \varepsilon_2, \varepsilon_1$. ** This list must contain exactly t_layers values.*
s_dielectric	List	A comma-separated list of dielectric constants for the layers below the TMD monolayer. Following the convention in Figure 2, the list is ordered starting from the layer closest to the TMD: s_dielectric= $\varepsilon_{\bar{1}}, \varepsilon_{\bar{2}}, \dots, \varepsilon_{\bar{s}}$. ** This list must contain exactly s_layers values.*
layer_boundaries	List	A comma-separated list of vertical coordinates (in Å), ordered from top to bottom, defining the planar interfaces between adjacent layers. **This list must contain exactly s + t values.*

Example File:

```
% Dielectric Function Control File

% --- Material Parameters for the TMD (SI units) ---

alpha = 1.55        % Dimensionless fitting parameter
Epl   = 22.5        % Plasmon peak energy (eV)
ep_p  = 13.24       % In-plane dielectric constant (static)

% --- Layer Architecture
% t: number of dielectric layers ABOVE the TMD
% s: number of dielectric layers BELOW the TMD
t_layers = 2
s_layers = 1

% --- Dielectric Constants of Surrounding Layers
% IMPORTANT:
% - Do NOT include the TMD monolayer here.
% Dielectric constants must be listed from top to bottom
t_dielectric = 1, 5.89
s_dielectric = 3.9

% --- Layer Boundaries (Top to Bottom)
% Units: Angstrom (Å)

layer_boundaries = 3.13, -3.13
```

3.2 User-Supplied Physical Data: User_input

The `User_input/` directory contains external physical data provided by the user to define the crystal geometry, electronic structure, dielectric screening, and real-space Wannier functions required for the calculations. Depending on the specific file type, ***WannierBSE*** manages these inputs using either a fallback priority sequence or strict consistency checks. For example, dielectric data follows a priority logic: the package first searches for a user-supplied file; if it is not found, the workflow falls back to an internal cache in `Precomputed_data/`; if neither is available, it dynamically generates the data. In contrast, structural and electronic inputs are tightly coupled to guarantee internal consistency across all workflows.

To construct the lattice vectors and atomic positions necessary for the tight-binding, k-mesh, dielectric, and exchange workflows, ***WannierBSE*** parses structural data from either `wannier90.win` or `structure.txt`. If both files coexist in the `User_input/` directory, the package prioritizes the Wannier90 input file (`wannier90.win`); otherwise, it reads the simplified `structure.txt`.

The requirements for user-supplied electronic data depend strictly on the chosen calculation mode:

- **Direct-Interaction-Only Calculations:** Users can supply an external electronic model, which must be provided as a complete, matching set consisting of `kmesh.txt` and a fully populated `TB_data/` directory. This directory must contain the corresponding valence (`v1_TB.txt` to `vN_v_TB.txt`) and conduction (`c1_TB.txt` to `cN_c_TB.txt`) band files. Providing an incomplete set or omitting either the k-mesh file or the band directory triggers an input-consistency error, halting execution. Once validated, ***WannierBSE*** automatically reorders and synchronizes the external band data to match its internal sorted k-point sequence. Alternatively, if an internal tight-binding model needs to be generated from scratch, the Wannier90 Hamiltonian file `wannier90_hr.dat` must be provided.
- **Direct-plus-Exchange Calculations:** External `kmesh.txt` and `TB_data/` models are not supported for calculations that include the electron-hole exchange interaction. Because the exchange kernel requires a consistent spinor Wannier basis, real-space spinor Wannier functions, and exchange tensors, these calculations must utilize the internal Wannier90-based workflow. Consequently, when exchange is enabled and the required preprocessed Wannier-function cache is not already available, users must provide the raw, real-space spinor Wannier functions within the `Wannier_functions_xsf/` directory. This directory must be organized into four mandatory component subfolders: `up_real/`, `up_imag/`, `down_real/`, and `down_imag/`. If the required Wannier-function cache or precomputed exchange tensors are already available and selected by the workflow, the raw XSF files do not need to be reprocessed.

Unlike the electronic structure restrictions, the dielectric input is decoupled from the calculation mode. In both direct-only and exchange-enabled simulations, users may optionally supply the dielectric screening function via `epsilon.txt`. If this file is missing, the package attempts to load the cached screening data

(`Precomputed_data/epsilon_WBSE.txt`). If no cache exists, it generates the dielectric data dynamically based on the parameters specified in `Parameters/dielectric_control.txt`.

Note: *This section strictly details the physical data explicitly provided by the user. Files generated, reordered, or cached automatically by the package are documented separately under the* `Precomputed_data/` *section.*

3.2.1 Momentum Space Grid: kmesh.txt

The file `User_input/kmesh.txt` defines the momentum-space grid used by an external electronic model. It specifies the k-points at which the user-supplied valence and conduction band energies, as well as the tight-binding eigenvectors, are defined. This file is central to the external electronic-model workflow and must always be accompanied by the corresponding band files in the `User_input/TB_data/` directory. Crucially, user-defined `kmesh.txt` files are applicable only when evaluating direct Coulomb interactions; exchange-enabled runs rely on Wannier90 Hamiltonians to compute spatial orbital overlaps.

The band files within `TB_data/` must be ordered consistently with the k-points listed in `kmesh.txt`. Specifically, each k-point block in every band file must strictly match the row order of the user-supplied k-mesh before ***WannierBSE*** performs its internal sorting and synchronization step.

During internal synchronization, ***WannierBSE*** first verifies that the total number of coordinate rows matches the sum of the dense and coarse point counts declared in the header. It then identifies the dense and coarse regions from the k-point spacing and sorts the dense-region points first, followed by the coarse-region points. The original `User_input/kmesh.txt` file is not modified. After a successful run, the final k-mesh used by the calculation is saved in `Exciton_data/k.mat`.

To be successfully parsed, the user-supplied `kmesh.txt` file must strictly follow a plain-text numeric table structure consisting of a five-line header containing mesh statistics, followed by the list of k-point coordinates

- Header Block (Lines 1–5):
    1. Dense Region Range: The size of the dense region in units of the coarse grid.
    2. Coarse Divisions: The number of divisions along the vector path in the coarse region.
    3. Dense Divisions: The number of divisions along the vector path in the dense region.
    4. Total Coarse Points: The total count of grid points belonging to the coarse region.
    5. Total Dense Points: The total count of grid points belonging to the dense region.

- Coordinate Block (Line 6 onward):
  The remaining lines list the Cartesian coordinates of every $\boldsymbol{k}$-point in the mesh. There are three columns corresponding to k_x, k_y, and k_z.

These coordinates must be provided in inverse Angstroms ($Å^{-1}$). ***WannierBSE*** currently utilizes the in-plane components kx and ky for the BSE workflow. Crucially, the file must contain only numeric entries in both the header and coordinate blocks. Do not include text, labels, or inline comments on any of the numeric lines.

Example file:

```
  10              % Dense Region Range
  20              % Coarse Divisions
  10              % Dense Divisions
1200              % Total Coarse Points
1200              % Total Dense Points
-0.3289873897   0.6078110657   0.0000000000
-0.2960886507   0.6647933531   0.0000000000
...
```

In this example, the header declares 1,200 coarse-region points and 1,200 dense-region points. Consequently, the coordinate block following line 5 must contain exactly 2,400 k-point rows.

3.2.2 Conduction and Valence Bands: TB_data/

The directory `User_input/TB_data/` contains the `c*_TB.txt` / `v*_TB.txt` files. Which stores the energy eigenvalues and the wavefunction coefficients (eigenvectors) in the Tight-Binding scheme for the specific bands involved in the BSE calculation. Importantly, user-supplied Tight-Binding files are supported strictly for calculations that consider direct interactions alone; they cannot be used when exchange interactions are enabled.

The filenames use a lowercase prefix followed by the integer band index. This index corresponds to the specific conduction or valence band sorted by energy distance from the gap (e.g., 1 is the band closest to the gap).

- c*_TB.txt: Conduction band data (e.g., `c1_TB.txt`).
- v*_TB.txt: Valence band data (e.g., `v1_TB.txt`).

Similar to the grid file, users may provide these files manually or allow ***WannierBSE*** to generate them via internal helper functions.

It is important to notice that, if provided manually, the data in these files must correspond exactly to the k-points defined in `kmesh.txt`. The number of blocks and the coordinate values must match line-by-line. However, as the k-mesh is automatically reorganized to prioritize the dense region, ***WannierBSE*** will dynamically rearrange the data blocks within the band files to ensure they are perfectly synchronized with the new momentum grid. This automated alignment maintains the physical integrity of the eigenvalues and eigenvectors without requiring the user to manually reorder the input files. For research traceability, these rearranged files are stored in the `TB_Bands/` folder within the `Exciton_data/` directory.

The file is structured as a sequence of data blocks. Each block corresponds to a single k-point in the mesh. Given a basis set of size N_{states}, each block contains $N_{states} + 2$ lines.

The block structure, repeated for every k-point, is as follows:

1. Line 1 (k -vector): The Cartesian coordinates k_x, k_y, k_z (in Å^{-1}).
2. Line 2 (Eigenvalue): The energy eigenvalue of the band ϵ_{nk} (in eV).
3. Lines 3 to End (Eigenvectors): The coefficients of the eigenvector expanded in the Wannier Tight-Binding basis. There are N_{states} lines. Each line contains three columns:

- $Re[C_j(k)]$: Real part of the coefficient.
- $Im[C_j(k)]$: Imaginary part of the coefficient.
- $Abs[C_j(k)]$: Absolute magnitude.

***Notation Alert**: Users should be careful to distinguish between the band index and the eigenvector coefficient index, as both typically use the letter 'c'. The lowercase notation c_n refers specifically to the n-th conduction band, which is reflected in the filename (e.g., `c1_TB.txt`). In contrast, the uppercase notation C_j refers to the j-th component of the eigenvector, represented by the individual rows of data within that file.*

Example file:

```
% --- Block for k_point 1 ---
 -0.328987   0.607811   0.000000    % k-vector (A^-1)
 -4.035566                          % Eigenvalue (eV)
 -0.161540  -0.166478   0.231970    % C_1 (Re, Im, Abs)
  0.004001   0.009269   0.010096    % C_2 (Re, Im, Abs)
 -0.223895  -0.030976   0.226027    % C_3 (Re, Im, Abs)
  ...
```

3.2.3 Dielectric Screening: epsilon.txt

This file contains the tabulation of the momentum-dependent static dielectric function, $\epsilon(|q|)$, which describes the screening of the Coulomb interaction in the 2D material system.

Users may choose to provide a pre-calculated dielectric function obtained from other methods or use the internal model provided by ***WannierBSE***. If the `epsilon.txt` file is not found in the `User_input/` directory, the package automatically invokes its internal Poisson solver to generate the screening data based on the layer architecture and material constants specified in `dielectric_control.txt` (see Section 3.1.6). Conversely, if the file is manually provided, the internal generation step is skipped, and `dielectric_control.txt` is ignored.

The file is formatted as a standard two-column ASCII text file. The first column represents the magnitude of the momentum transfer vector, $|q|$, in units of $Å^{-1}$, covering the range $0 \leq |q| \leq \infty$. The second column contains the corresponding dimensionless dielectric function value, $\epsilon(|q|)$.

Example file:

```
%---       q(A⁻¹)          epsilon(q)
       0.000000e+00      1.000000e+00
       2.500250e-04      1.009815e+00
       5.000500e-04      1.019619e+00
       7.500750e-04      1.029413e+00
       1.000100e-03      1.039197e+00
       1.250125e-03      1.048970e+00
       ...
```

3.2.4 Hamiltonian Matrix: wannier90_hr.dat

This file contains the Hamiltonian matrix elements calculated in the maximally localized Wannier function basis. It is a standard output generated by the Wannier90 software package.

Within the ***WannierBSE*** workflow, this file serves as the fundamental source data. If the processed wavefunction files (`c*_TB.txt` and `v*_TB.txt`) are not present in the input directory, the code reads `wannier90_hr.dat` to construct the Hamiltonian, diagonalize it at the required k-points, and generate the missing wavefunction files automatically. Users should simply copy this file from their Wannier90 output folder into the `User_input/` directory without modification.

The file follows the standard format defined by the Wannier90 developers, consisting of a header defining the system size followed by the list of matrix elements. For a detailed line-by-line description of the headers, degeneracy blocks, and column definitions, please refer to the official Wannier90 User Guide.

Example File:

```
written on 14 Jul 2025 at 10:15:35
      22
      91
3    2    1    1    2    2    1    1    1    1    1    2    3    1    1
1    1    1    1    1    1    3    1    1    1    1    1    1    1    1
...
   -6    3    0    1    1    0.000236   -0.000000
```

```
 -6    3    0    2    1     0.000000     0.000000
 -6    3    0    3    1     0.000000     0.000000
 -6    3    0    4    1     0.000005     0.000000
 -6    3    0    5    1     0.000005    -0.000000
 -6    3    0    6    1    -0.000000     0.000005
 -6    3    0    7    1    -0.000270     0.000000
 ...
```

3.2.5 Structure Definition for Model-Based Systems: structure.txt

The `structure.txt` file defines the crystal lattice vectors and atomic positions specifically for custom model-based or tight-binding calculations. However, this file mode is only supported when computing direct electron-hole interactions. When the exchange interaction is included, ***WannierBSE*** strictly requires the system structure to be read from `wannier90.win` instead (Section 3.2.6).

This design choice ensures input consistency across the entire calculation pipeline. To maintain a unified workflow and avoid redundancy, users are strongly encouraged to supply `wannier90.win` directly for all standard Wannier90-based runs, regardless of whether the exchange interaction is active.

The format strictly follows the conventions used by the Wannier90 software package, facilitating easy migration of existing input data.

The lattice vectors are specified in Cartesian coordinates enclosed within the *unit_cell_cart* block. The components must be provided in Angstroms (Å). The block format is as follows:

begin unit_cell_cart

$$
\begin{array}{ccc}
A_x^1 & A_y^1 & A_z^1 \\
A_x^2 & A_y^2 & A_z^2 \\
A_x^3 & A_y^3 & A_z^3
\end{array}
$$

end unit_cell_cart

In this configuration, the values A_x^1, A_y^1, A_z^1 correspond to the Cartesian components of the first lattice vector A_1 with subsequent rows corresponding to vectors A_2 and A_3, respectively.

Similarly, the atomic positions are defined using the atoms_cart block. These coordinates must also be specified in Angstroms (Å) and absolute Cartesian coordinates.

begin atoms_cart

Label R_x R_y R_z

...

end atoms_cart

The first entry on each line, *Label*, represents the atomic symbol (e.g., Mo, S). The following three entries $R = (R_x, R_y, R_z)$ denote the position coordinates of that atom in the Cartesian frame.

Example File:

```
begin unit_cell_cart
  2.7566392707   0.0000000000   0.0000000000
 -1.5915472948   2.7566392707   0.0000000000
  0.0000000000   0.0000000000  15.1264784281
end unit_cell_cart

begin atoms_cart
Mo   0.0000000000   3.6755187193   7.5632392140
S    0.0000000000   1.8377596680   9.1267531759
S    0.0000000000   1.8377596680   5.9997252522
end atoms_cart
```

3.2.6 Structure Definition for Full Wannier-Based Systems: wannier90.win

This file is the master input file used by the Wannier90 software package, specifying the crystal geometry, unit cell, and system configuration.

Within the ***WannierBSE*** workflow, this file serves as the primary source for defining the crystal lattice vectors and atomic positions. It is the standard structure input for all Wannier90-based calculations and is mandatory whenever the exchange interaction is included, ensuring consistency across both direct and exchange interaction modes. When performing calculations that involve exchange, the code reads `wannier90.win` directly to extract structural information alongside the electronic matrix elements. Users should simply copy or link this file from their original Wannier90 input folder into the `User_input/` directory without modification.

The file follows the standard format defined by the Wannier90 developers, consisting of keyword blocks such as unit_cell_cart (or unit_cell_frac) and atoms_cart (or atoms_frac). For a detailed description of the keywords, syntax rules, and unit specifications, please refer to the official Wannier90 User Guide.

3.2.7 Spinor Wannier Function: wannier90_*.xsf

The folder `User_input/Wannier_functions_xsf/` contains the real-space Wannier functions exported by the Wannier90 package in XSF format. These files are used as the starting point for constructing the real-space spinor Wannier basis used in the exchange calculation. Because the calculation uses spinor Wannier functions, each Wannier function has two spin components, up and down, and each spin component is complex.

Therefore, each Wannier function is represented by four XSF files:

- up_real.xsf: Real part of the spin-up component.
- up_imag.xsf: Imaginary part of the spin-up component.
- down_real.xsf: Real part of the spin-down component.
- down_imag: Imaginary part of the spin-down component.

Thus, the four files do not correspond to four different Wannier functions. They are the four components needed to reconstruct one complex spinor Wannier function. For the XSF format: All XSF files have the same general format. Each file contains:

1. A header indicating that the file was generated by Wannier90.
2. The crystal lattice vectors, given in the PRIMVEC block.
3. The atomic positions, given in the PRIMCOORD block.
4. A three-dimensional real-space grid, given in the BEGIN_DATAGRID_3D block.
5. The complex scalar value of one Wannier-function component on each grid point.

For a more detailed line-by-line description please refer to the official Wannier90 User Guide.

Example File:

```
#
# Generated by the Wannier90 code http://www.wannier.org
# On 27Jan2026 at 21:47:43

CRYSTAL
PRIMVEC       # Primitive lattice vectors
   1.5915473   2.7566393   0.0000000
  -1.5915473   2.7566393   0.0000000
   0.0000000   0.0000000  15.1264784
CONVVEC       # Conventional lattice vectors
   1.5915473   2.7566393   0.0000000
  -1.5915473   2.7566393   0.0000000
   0.0000000   0.0000000  15.1264784
PRIMCOORD
     3  1     # Number of atoms and coordinate type
Mo       0.0000000   3.6755187   7.5632392 # Atom Cartesian position
S        0.0000000   1.8377597   9.1267532
S        0.0000000   1.8377597   5.9997253


BEGIN_BLOCK_DATAGRID_3D
3D_field
BEGIN_DATAGRID_3D_UNKNOWN
   216   216    90      # Number of grid points along x, y, and z
    0.000000  -33.385965   -0.168072 # Datagrid origin
  19.0101482  32.9265246   0.0000000 # 3D datagrid vectors
 -19.0101482  32.9265246   0.0000000
   0.0000000   0.0000000  14.9584064
 0.106E-06 -0.110E-06  0.559E-07  0.179E-07 -0.864E-07  0.597E-07
-0.106E-06  0.175E-07 -0.283E-07 -0.949E-07  0.384E-07 -0.877E-07
...
END_DATAGRID_3D
END_BLOCK_DATAGRID_3D
```

3.3 Physical Data: Precomputed_data

The `Precomputed_data/` directory stores the WBSE-ready physical data that ***WannierBSE*** generates, preprocesses, and reuses during calculations. While `User_input/` contains physical data supplied directly by the user, `Precomputed_data/` contains the internally consistent versions used by the solver, together with cached intermediate data that can accelerate subsequent runs.

When a user-supplied file and its corresponding precomputed file are both available, ***WannierBSE*** gives priority to the file in `User_input/`. For example, `User_input/kmesh.txt` and `User_input/epsilon.txt` are used before `Precomputed_data/kmesh_WBSE.txt` and `Precomputed_data/epsilon_WBSE.txt`. When needed, the user-provided data are validated, sorted, synchronized, and written into `Precomputed_data/` so that the remaining calculation proceeds from a consistent internal representation.

If the required file is not provided in `User_input/`, ***WannierBSE*** searches for the corresponding cached file in `Precomputed_data/`. When the cache is complete and compatible with the requested calculation, it is loaded directly. Otherwise, ***WannierBSE*** regenerates the required data from the control parameters and Wannier90 inputs, then stores the result in `Precomputed_data/` for later reuse.

In addition to the WBSE momentum grid `kmesh_WBSE.txt`, dielectric function `epsilon_WBSE.txt`, and tight-binding band files stored in TB_data/, this directory may contain the reordered Wannier-center file `WF_centers_WBSE.txt`, preprocessed Wannier-function data in `Wannier_functions_WBSE/`, and reusable exchange-interaction tensors in `Exchange_interaction/`. These files are described individually in the following subsections.

The files `kmesh_WBSE.txt`, `epsilon_WBSE.txt`, `v*_TB_WBSE.txt`, and `c*_TB_WBSE.txt` follow the same structure as their corresponding user-supplied files without the _WBSE suffix. Their formats are therefore not repeated here; instead, the correspondence between the `User_input/` and `Precomputed_data/` versions is summarized in the table below.

Data Type	**User Input Name**	**Automated Cache Name (Precomputed_data/)**
K-Mesh	`kmesh.txt`	kmesh_WBSE.txt
Bands	`v*_TB.txt,` `c*_TB.txt`	`v*_TB_WBSE.txt,` `c*_TB_WBSE.txt`
Dielectric	`epsilon.txt`	`epsilon_WBSE.txt`

Note on File Formats: *The internal structure (columns and units) of these files is identical regardless of their location. For example, a v1_TB.txt provided by the user must follow the same column specifications as a generated v1_TB_WBSE.txt file. For detailed file structure specifications, refer to section 3.2.*

3.3.1 Spin-Block-Ordered Wannier Centers: WF_centers_WBSE.txt

The `WF_centers_WBSE.txt` file is the internal ***WannierBSE*** adaptation of `Parameters/WF_centers.txt`. It retains the core physical data, Wannier-function index, associated atomic center, and spinor component, while reorganizing the entries into a spin-block convention, grouping all spin-up functions prior to spin-down functions.

This reordered basis ensures structural consistency across Wannier centers, tight-binding eigenvector coefficients, and short-range exchange matrix elements within the internal Wannier90-based pipeline.

Example File:

```
    i      I      Spinor
```

```
   1      2      up
   2      2      up
   3      2      up
   4      3      up
...
  12      2      down
  13      2      down
  14      2      down
  15      3      down
```

3.3.2 Processed Wannier Function: Wannier_functions_WBSE/

The `Precomputed_data/Wannier_functions_WBSE/` directory serves as an internal cache storing processed real-space spinor Wannier functions. This directory is generated only when exchange interactions are enabled (*Exchange_Interaction* = *true*); it is omitted during direct-interaction-only calculations.

During execution, ***WannierBSE*** checks this internal cache before processing raw volumetric XSF files from `User_input/Wannier_functions_xsf/`. If cached files are present, they are loaded directly. If missing, the software parses the XSF files, shifts and normalizes the wavefunctions, and converts them into compact MATLAB binary (.mat) files formatted in the internal basis convention. This conversion significantly reduces disk storage requirements and speeds up data loading during exchange kernel construction.

If *cleanup_xsf* = *true* is specified in `exchange_control.txt`, ***WannierBSE*** automatically deletes the raw input XSF files after conversion while maintaining the processed binary cache for future runs.

All files within `Wannier_functions/` form an interdependent dataset that must be maintained as a complete set. The spatial evaluation grid, atomic coordinate metadata, and spinor wavefunction matrices all share identical grid dimensions and basis ordering. Wavefunctions are ordered according to the spin-block convention in `Precomputed_data/WF_centers_WBSE.txt` (spin-up components grouped first, followed by spin-down components), ensuring precise spatial mapping of spinor amplitudes during exchange-tensor assembly.

The standard files contained in this directory include:

- `GridInfo.txt`: Plain-text metadata specifying grid dimensions, origin coordinates, and lattice vectors.
- `r.mat`: Binary matrix containing the Cartesian coordinates of the real-space evaluation grid.
- `WF_AtomPosition.txt`: Atomic-position coordinates aligned with the Wannier-function real-space grid.

- `WF_up.mat`: Processed binary data for the spin-up components of the Wannier wavefunctions.
- `WF_down.mat`: Processed binary data for the spin-down components of the Wannier wavefunctions.

Caution: *Manual editing or partial replacement of files in this directory is strongly discouraged. Because the dataset is strictly coupled, modifying any single file without regenerating the full set breaks spatial alignment between coordinates, spinor components, and Wannier basis functions, leading to unphysical results or execution failure.*

3.3.2.1 Real-Space Grid Metadata: GridInfo.txt

The `GridInfo.txt` file specifies the spatial sampling parameters of the real-space grid used for the processed Wannier functions. ***WannierBSE*** generates this metadata file during preprocessing by extracting grid parameters from the input XSF files and storing it alongside `r.mat`, `WF_up.mat`, and `WF_down.mat` in `Precomputed_data/Wannier_functions_WBSE/`. During exchange-enabled calculations, the solver reads `GridInfo.txt` to determine grid dimensions, differential integration volume elements, supercell boundaries, and origin translation vectors essential for evaluating real-space exchange integrals.

The file is a plain-text table consisting of five rows, with each row containing three numeric entries. Row 1 specifies integer grid divisions, while Rows 2–5 define Cartesian spatial vectors (typically in Angstroms (Å)):

- Line 1 ($N_1^a \;\; N_2^a \;\; N_3^a$): Number of real-space grid sampling points along the three lattice directions.
- Line 2 ($O_x \;\; O_y \;\; O_z$): Cartesian coordinates of the grid origin.
- Line 3 ($A_x^1 \;\; A_y^1 \;\; A_z^1$): Cartesian components of the first supercell vector.
- Line 4 ($A_x^2 \;\; A_y^2 \;\; A_z^2$): Cartesian components of the second supercell vector.
- Line 5 ($A_x^3 \;\; A_y^3 \;\; A_z^3$): Cartesian components of the third supercell vector.

Extracted directly from the header block of the input XSF files, these parameters preserve the precise grid geometry of the original sampling mesh. The total number of spatial grid points, given by the product $N_{grid} = N_{a1} \times N_{a2} \times N_{a3}$, must strictly equal the row dimension of `r.mat`, `WF_up.mat`, and `WF_down.mat`.

Example File:

```
 216   216   90          % Na1, Na2, Na3 (Total points: 216*216*90)
  0.000 -33.385  -0.168 % Origin (Ox, Oy, Oz)
 19.010  32.926   0.000 % First supercell vector A1
-19.010  32.926   0.000 % Second supercell vector A2
  0.000   0.000  14.958 % Third supercell vector A3
```

3.3.2.2 Real-Space Coordinate Grid: r.mat

The `r.mat` file stores the Cartesian real-space evaluation grid used to represent the processed spinor Wannier functions in `WF_up.mat` and `WF_down.mat`. ***WannierBSE*** automatically generates this file during preprocessing after parsing the raw XSF coordinate grid and applying internal spatial shifts and ordering conventions. During exchange-enabled calculations, the solver loads `r.mat` to map coordinates into the working lattice frame, isolate localized Wannier regions, and compute real-space exchange integrals.

The file is a binary MATLAB data file containing a single double-precision matrix named r of dimensions $N_{grid} \times 3$, where N_{grid} represents the total number of spatial grid points. Each row defines a spatial coordinate with the following column structure:

- $r(:1)$: Cartesian x-coordinate.
- $r(:2)$: Cartesian y-coordinate.
- $r(:3)$: Cartesian z-coordinate.

Strict row alignment is maintained across all real-space binary files: row $\underline{i}$ in `r.mat` corresponds directly to row $\underline{i}$ in both `WF_up.mat` and `WF_down.mat`. Consequently, spinor wavefunction amplitudes and spatial coordinates must always be evaluated using the same row index. Spatial units inherit the convention of the input XSF dataset, conventionally given in units of Angstroms (Å).

Example File:

```
%--   rx            ry            rz

 rx_1          ry_1          rz_1          % Grid point 1 (x, y, z)
 rx_2          ry_2          rz_2          % Grid point 2 (x, y, z)
 rx_3          ry_3          rz_3          % Grid point 3 (x, y, z)
 ...           ...           ...
 rx_Ngrid      ry_Ngrid      rz_Ngrid      % Grid point N_grid (x, y, z)
```

3.3.2.3 Wannier Grid Atom Positions: WF_AtomPosition.txt

The `WF_AtomPosition.txt` file records the Cartesian origin and atomic positions associated with the real-space Wannier-function grid. During preprocessing, ***WannierBSE*** extracts these geometric parameters directly from the metadata block of the reference spinor-resolved XSF files, storing the resulting file in `Precomputed_data/Wannier_functions_WBSE/` alongside `r.mat`, `GridInfo.txt`, `WF_up.mat`, and `WF_down.mat`. This file provides structural traceability for the cached real-space dataset across subsequent calculation steps.

The file is a plain-text table with three numeric columns. Line 1 specifies the Cartesian grid origin, while subsequent lines define the Cartesian coordinates of each atom in the XSF structure block:

- Line 1 (O_x O_y O_z): Cartesian origin coordinates of the Wannier-function evaluation grid.
- Line 2 to $N_{atoms} + 1$ (X_i Y_i Z_i): Cartesian position coordinates for the i^{th}-atom with $i = 1, \cdots, N_{atoms}$.

Spatial units inherit the input XSF convention, measured in Angstroms (Å). The ordering of atomic rows must strictly align with the atomic sequence defined in the main structural input and `Parameters/WF_centers.txt`, as Wannier-center mapping relies directly on these row indices.

Example File:

```
0.000 -33.385 -0.168 % Grid Origin (Ox, Oy, Oz)
0.000   3.675  7.563 % Atom 1 position (X1, Y1, Z1)
0.000   1.837  9.126 % Atom 2 position (X2, Y2, Z2)
0.000   1.837  5.999 % Atom 3 position (X3, Y3, Z3)
```

3.3.2.4 Binary Spinor Wannier Functions: WF_down.mat/WF_up.mat

The `WF_up.mat` and `WF_down.mat` files store the processed real-space spinor Wannier functions required by the exchange interaction workflow. During preprocessing, ***WannierBSE*** parses the real and imaginary volumetric XSF data from `User_input/Wannier_functions_xsf/`, applies spatial grid shifts, normalizes the wavefunctions, reorders the Wannier basis into the internal spin-block convention, and exports the resulting complex amplitudes into compact MATLAB binary format. During exchange-enabled calculations, the solver loads these binary files alongside `r.mat` and `GridInfo.txt` to isolate localized wavefunction regions and evaluate real-space overlap integrals for exchange-tensor assembly.

Each file is a MATLAB binary file containing one complex numeric matrix: `WF_up.mat` contains the up-spinor component of the processed Wannier functions, while `WF_down.mat` contains the down-spinor component. Both matrices (WF_up and WF_down) share matching dimensions of $N_{grid} \times N_{WF}$. Matrix rows correspond to spatial grid points indexed by $i = 1, \cdots, N_{grid}$, maintaining strict row alignment with row i of `r.mat`. Matrix columns correspond to Wannier basis functions indexed by $j = 1, \cdots, N_{WF}$, which follow the exact spin-block ordering specified in `Precomputed_data/WF_centers_WBSE.txt`. Together, the pair of column vectors WF_up(:,j) and WF_down(:,j) defines the complete two-component spinor wavefunction for Wannier function j across the full spatial mesh:

- WF_up(i,j): Complex spin-up amplitude of Wannier function j evaluated at spatial grid point i.
- WF_down(i,j): Complex spin-down amplitude of Wannier function j evaluated at spatial grid point i.

Because these binary cache files can be quite large, full console printing is discouraged; verification should be performed using matrix dimension checks or localized diagnostics in MATLAB.

Example File:

```
%---  WF_up

%     WF_1          WF_2      ...     WF_NWF
  WF_up_1(r1)   WF_up_2(r1)   ... WF_up_NWF(r1)   % Grid point 1
  WF_up_1(r2)   WF_up_2(r2)   ... WF_up_NWF(r2)   % Grid point 2
  WF_up_1(r3)   WF_up_2(r3)   ... WF_up_NWF(r3)   % Grid point 3
  ...
  WF_up_1(rN)   WF_up_2(rN)   ... WF_up_NWF(rN)   % Grid point N_grid
```

```
%---  WF_down

%     WF_1           WF_2      ...     WF_NWF
  WF_down_1(r1)   WF_down_2(r1)   ... WF_down_NWF(r1) % Grid point 1
  WF_down_1(r2)   WF_down_2(r2)   ... WF_down_NWF(r2) % Grid point 2
  WF_down_1(r3)   WF_down_2(r3)   ... WF_down_NWF(r3) % Grid point 3
  ...
  WF_down_1(rN)   WF_down_2(rN)   ... WF_down_NWF(rN) % Grid point N_grid
```

3.3.3 Precomputed Exchange Interaction Cache: Exchange interaction/

The `Precomputed_data/Exchange_interaction/` directory stores the reusable exchange-interaction cache generated by ***WannierBSE***. This folder contains the retained lattice-translation dataset, `R_mNN_WBSE.mat`, and the corresponding exchange tensor, `chi_WBSE.mat`. These cache files are produced only when exchange interactions are enabled, *Exchange_Interaction = true*, and precomputed data is not being explicitly loaded, *Use_Precomputed_Chi = false*.

When both *Exchange_Interaction = true* and *Use_Precomputed_Chi = true* are specified in `exchange_control.txt`, ***WannierBSE*** bypasses the computationally demanding real-space exchange integration step. Instead, the solver directly loads `R_mNN_WBSE.mat` and `chi_WBSE.mat` from this directory, verifies variable existence and dimension compatibility across the retained lattice vectors, and integrates the cached exchange tensors into the BSE Hamiltonian assembly. Reusing this cache provides significant computational speedups, as evaluating exchange integrals is typically the most time-consuming stage of the preprocessing workflow.

The exchange cache is fully decoupled from both the dielectric screening model, epsilon_WBSE.txt, and the BSE momentum mesh, kmesh_WBSE.txt. Consequently, a single pair of R_mNN_WBSE.mat and chi_WBSE.mat files can be reused across different dielectric environments, substrate screening configurations, or k-mesh convergence tests, provided the underlying Wannier basis and atomic geometry remain unchanged.

It is important to note that the exchange cache is not universal and must be regenerated whenever there are modifications to the crystal structure, input Wannier functions, basis ordering, WF_centers.txt, or exchange cutoff parameters in exchange_control.txt (such as M_max, G_max, tol_probability, or neighbor tolerances). While ***WannierBSE*** performs automated matrix-dimension and variable sanity checks upon loading, users are responsible for verifying that cached files match the physical system and parameters under study.

The primary files stored in this directory include:

- R_mNN_WBSE.mat: Retained real-space lattice translations utilized by the exchange interaction kernel.
- chi_WBSE.mat: Precomputed exchange tensor evaluated from the processed spinor Wannier functions and retained lattice vectors.

3.3.3.1 Retained Exchange Translation Vectors: R_mNN_WBSE.mat

The R_mNN_WBSE.mat file stores the set of real-space lattice translation vectors retained for the exchange interaction calculation. ***WannierBSE*** generates this binary file during the exchange preprocessing stage using the unit cell vectors, atomic positions, supercell expansion bounds, and nearest-neighbor cutoff criteria configured in Parameters/exchange_control.txt. These retained lattice vectors define the real-space displacement framework utilized alongside chi_WBSE.mat to evaluate phase factors during BSE Hamiltonian construction.

The spatial extent of R_mNN_WBSE.mat is governed by the user-defined M_max parameter in exchange_control.txt, which dictates the maximum nearest-neighbor shell cutoff for short-range exchange integrals. Increasing M_max expands the retained translation set, with the precise vector count determined by the crystal lattice geometry, atomic basis positions, supercell size, and distance classification tolerances (nn_distance_tolerance). Each retained translation vector R_m is constructed from integer lattice multipliers $n_{1,m}^{(R)}$ and $n_{2,m}^{(R)}$ and the primitive in-plane lattice vectors a_1 and a_2 ,

$$R_m = n_{1,m}^{(R)}\, a_1 + n_{2,m}^{(R)} a_2, m = 1, \cdots, N_R$$

where m indexes the translation vector and N_R denotes the total number of retained translation vectors in the set. For localized orbital pairs at atomic basis sites τ_i and τ_j, the relative real-space displacement vector evaluated against the exchange cutoff radius is:

$$\Delta r_{ij}^{m} = R_m + (\tau_j - \tau_i)$$

Only translation vectors meeting the distance criteria across atomic pairs are preserved in the binary cache.

The file contains a single real double-precision matrix named R_mNN of dimensions $N_R \times 3$, where rows correspond to discrete translation indices m and columns represent Cartesian vector components in Angstroms (Å):

- R_mNN(:,1): Cartesian x-component of the retained lattice translation vector.
- R_mNN(:,2): Cartesian y-component of the retained lattice translation vector.
- R_mNN(:,3): Cartesian z-component of the retained lattice translation vector.

Row indexing in R_mNN is strictly synchronized with the translation vector dimensions of the exchange tensor stored in `chi_WBSE.mat`. ***WannierBSE*** validates this cross-file index alignment automatically whenever precomputed exchange caches are loaded.

Example File:

```
%---   R_mNN

%               Rx          Ry         Rz
           -1.5915    -2.7566       0       % R1=(-a1,0,0)
            1.5915    -2.7566       0       % R2=(0,-a2,0)
            0           0           0       % R3=(0,0,0)
           -1.5915     2.7566       0       % R4=(0,a2,0)
            1.5915     2.7566       0       % R5=(a1,0,0)
```

3.3.3.2 Precomputed Exchange Tensor: chi_WBSE.mat

The `chi_WBSE.mat` file stores the precomputed real-space exchange tensor $\chi_{j'jl'l}^{RR'}(k_{ex})$ utilized when constructing the exchange contribution to the BSE Hamiltonian (see Equation 2.7) ***WannierBSE*** evaluates this tensor during the exchange preprocessing stage by integrating the processed spinor Wannier functions across the retained lattice translation vectors in `R_mNN_WBSE.mat` alongside reciprocal lattice parameters. When both *Exchange_Interaction = true* and *Use_Precomputed_Chi = true* are set in `exchange_control.txt`, the solver loads this tensor directly into memory, bypassing the computationally intensive real-space integration pipeline.

The file is a binary MATLAB data file containing a single complex double-precision tensor named chi. The chi array is six-dimensional, with array bounds structured as chi(N_{WF}, N_{WF}, N_{WF},N_{WF}, N_R, N_R):

- Dimension 1: First Wannier basis function index $j' = 1, \cdots, N_{WF}$.

- Dimension 2: Second Wannier basis function index $j = 1, \cdots, N_{WF}$.
- Dimension 3: Third Wannier basis function index $l' = 1, \cdots, N_{WF}$.
- Dimension 4: Fourth Wannier basis function index $l = 1, \cdots, N_{WF}$.
- Dimension 5: First retained lattice translation index $R = R_1, \cdots, R_{N_R}$.
- Dimension 6: Second retained lattice translation index $R' = R'_1, \cdots, R'_{N_R}$.

Indices *1* through *4* run over the Wannier basis ordered according to the spin-block convention in `Precomputed_data/WF_centers_WBSE.txt`. Dimensions *5* and *6* correspond strictly to the retained real-space translation vectors stored in `R_mNN_WBSE.mat`. Consequently, the translation dimension N_R in chi must match the row count of R_mNN exactly; ***WannierBSE*** verifies this dimension compatibility upon importing the tensor cache.

3.4 Output Files: Exciton_Data

This section describes the output files generated by the ***WannierBSE*** package. After the input configuration is finalized, the simulation is initiated by executing the `WBSE.m` main script. Upon successful completion, the package produces a suite of files containing the primary physical results and auxiliary simulation data.

The primary physical results are contained in two main files: `Ex.mat`, which stores the exciton energy spectrum, and `A.mat`, which contains the excitonic wavefunctions. In addition to these core results, the package generates three auxiliary log files, `WBSE_log.txt`, `k.mat`, and `wi.mat`, that record the simulation performance and the specific grid geometry used.

When the exchange workflow is active and the exchange tensors are computed during the run, ***WannierBSE*** also creates the `Exchange_data/` directory. This folder stores run-level copies of exchange-related intermediate data including `rho.mat`, `G_all.mat` and files like `chi.mat` and `R_mNN.mat`, which have the same structure than `chi_WBSE.mat` and `R_mNN_WBSE.mat` (see section 3.3.3). These files provide traceability for the exchange calculation associated with the current output dataset.

To ensure efficient storage and fast post-processing, the physical data is saved in MATLAB binary format (.mat). Detailed descriptions of the format and content of each file are provided in the following subsections.

3.4.1 Exciton Energy: Ex.mat

This file represents one of the main outputs of the ***WannierBSE*** package. It stores the fundamental results of the BSE diagonalization: the exciton energy eigenvalues, denoted as E_S^X.

The data is stored as a double-precision column vector. The eigenvalues are arranged in ascending order (from lowest energy to highest). The total number of entries corresponds to the parameter NVAL specified in the `control.txt` file. All energy values are provided in electron-volts (eV).

Example file:

```
%---              E_S^X
           1.1180   % E_1^X
           1.1187   % E_2^X
           1.1455   % E_3^X
           1.1455   % E_4^X
           1.2652   % E_5^X
             ...
```

3.4.2 Exciton Wavefunctions: A.mat

Together with `Ex.mat`, this file constitutes the core output of the ***WannierBSE*** package. It contains the excitonic envelope functions (eigenvectors), denoted as $A_S(vck)$, which describe the probability amplitude of the electron-hole pair in reciprocal space.

The data is stored as a 2D matrix of complex double-precision numbers.

- Columns (Exciton States): Each column corresponds to a specific exciton eigenstate S, sorted by energy in ascending order.
- Rows (Basis Index): This index is defined by the specific combination of the valence band index (v_i), the conduction band index (c_j), and the momentum vector (k_l).

The elements of the eigenvector are organized hierarchically. The file lists the coefficients for all k-points for the first band pair (v_1, c_1), followed by the next conduction band (v_1, c_2), continuing until all conduction bands are exhausted for the first valence band. This entire sequence repeats for the next valence band (v_2), and so on, until all valence-conduction combinations are covered.

The total number of rows corresponds to the size of the basis set:

$$Total\ Rows = N_v \times N_c \times Total\ k\ points$$

Example file:

```
%--   Exciton State S=1        Exciton State S=2     ...   Index
    1.308e-07 -9.886e-08i    1.381e-06 -1.138e-07i   ...  %(v1c1k1)
    2.919e-08 -8.831e-08i    2.418e-06 -2.237e-05i   ...  %(v1c1k2)
    ...
    5.318e-03 -6.866e-07i    7.937e-01 +4.489e-06i   ...  %(v1c2k1)
    ...
    3.450e-07+ 6.726e-08i    4.381e-06 -1.138e-03i   ...  %(vNvcNck1)
    ...
```

3.4.3 Simulation Log: WBSE_log.txt

WBSE_log.txt is a plain-text execution report generated at the end of every successful ***WannierBSE*** run. It provides a compact summary of the calculation for performance tracking, reproducibility checks, and benchmarking across different physical or numerical configurations. The log details key runtime parameters, including the BSE basis dimensions, Coulomb-interaction parameters, parallel worker distributions, and stage-by-stage wall-clock timings.

The log is organized into four sections: BSE BASIS, COULOMB INTERACTION, CPU USAGE, and COMPUTATION TIME. The first two sections document the active k-point grid, band indices, and exchange settings (such as M_max, G_max, and associated tolerance thresholds), indicating whether exchange tensors were computed on the fly or loaded from precomputed data. The remaining sections detail the parallel resources assigned to each workflow stage, flagging unused stages as not used, and report measured wall-clock execution times for matrix construction, diagonalization, and optional preprocessing steps like exchange-tensor generation.

Example file:

```
WannierBSE calculation report
Generated at: 2026-07-28 13:29:50

==================== BSE BASIS ====================
Number of k-points                       = Nk = 1350
Valence bands used                       = Nv = 2
Conduction bands used                    = Nc = 2

==================== COULOMB INTERACTION ====================
Short-range exchange interaction         = enabled
Exchange tensor source                   = computed during current run

Nearest-neighbor Wannier shell         M_max               = 1
Largest short-range G-vector cutoff    G_max               = 4.2
Wannier truncation probability cutoff  tol_probability     = 0.98
M-neighbor distance tolerance          M_max_tolerance     = 0.01
Nearest-neighbor distance tolerance    nn_distance_tolerance = 0.15

==================== CPU USAGE ====================
TB calculations                      Npar_TB                = 12
rho calculation                      Npar_rho               = 40
Exchange tensor calculation          Npar_Xi                = 40
BSE Hamiltonian construction         Npar_HBSE              = 40

==================== COMPUTATION TIME ====================
rho overlap-density calculation               =      468.180 sec
short-range exchange tensor calculation       =      514.819 sec
BSE Hamiltonian matrix construction           =      196.198 sec
BSE Hamiltonian diagonalization               =       44.816 sec
Total BSE solver stage                        =      241.014 sec
Total WBSE execution                          =     3124.720 sec
```

3.4.4 Momentum Grid Data: k.mat

This MATLAB binary file stores the explicit coordinates of the momentum space grid used in the calculation. It contains the exact same vector data as the text-based kmesh.txt file but is saved in a binary format to facilitate direct and efficient loading during post-processing. The data is stored as a double-precision matrix with dimensions $N_k \times 3$. Each row corresponds to a unique k-point index i. The three columns represent the Cartesian componentsk_x, k_y, and k_z (in $Å^{-1}$). Consistent with the validation and sorting logic described in Section 3.2.1, the grid points in this file are organized so that the dense region is listed first, followed by the coarse region. For detailed information regarding the grid generation logic and the distribution of points, please refer to the Momentum Space Grid (`kmesh.txt`) description in Section 3.3.1.

Example file:

```
      kx             ky             kz
-0.3289873897   0.6078110657   0.0000000000  % k1
-0.2960886507   0.6647933531   0.0000000000  % k2
-0.3289873897   0.6457992573   0.0000000000  % k3
     ...            ...            ...
```

3.4.5 Integration Weights: wi.mat

This MATLAB binary file stores the integration weights (w_i) associated with the momentum grid. These weights are mathematically derived from the discretization of the Bethe-Salpeter integral equation into a matrix eigenvalue problem. In the numerical implementation, the continuous integration over the Brillouin zone is replaced by a summation over discrete k-points:

$$\int \frac{d^2k}{(2\pi)^2} \cdots \to \sum_i \frac{w_i}{(2\pi)^2} \cdots$$

Geometrically, each k-point in the mesh represents the centroid of a specific area element, typically a small triangle within the D_{3h} symmetry mesh. The value w_i corresponds to the area of this Voronoi cell or triangle occupied by the i-th k -point. These weights are essential for properly normalizing the exciton wavefunctions and for computing optical matrix elements.

Regarding the file structure, the data is stored as a double-precision column vector with dimensions $total\ number\ of\ k-points \times\ 1$. The ordering of the rows strictly corresponds to the sequence of k -points listed in `k.mat` and `kmesh.txt`, such that the value in the i-th row represents the specific weight for the coordinate found in the i-th row of the grid files.

Example file:

```
0.001874645399843  % w1
0.001874645399843  % w2
0.001874645399843  % w3
0.001874645399843  % w4
0.001874645399843  % w5
        ...
```

3.4.6 Rearranged Band Data: TB_Bands/

This sub-directory is a specialized repository used for research traceability. It is generated only when a user manually provides conduction or valence band files (`c*_TB.txt` or `v*_TB.txt`) in the User_input directory.

As discussed in Section 3.2.2, the software automatically reorganizes manual input files to ensure the k-point sequence matches the internal dense-to-coarse sorting required for numerical stability. To ensure the calculation remains transparent, ***WannierBSE*** saves the resulting "synchronized" versions of these files here. It is important to note that the internal structure of these files, including the format for eigenvalues and wavefunction coefficients, remains exactly as detailed in Section 3.2.2.

- Content: Contains rearranged versions of the user-provided .txt band files.
- Condition: This folder will not be created if the software generates the bands internally, as those results are stored in the `Precomputed_data/` folder.
- Usage: Users can refer to these files to verify the exact alignment of eigenvalues and eigenvectors used in the final BSE diagonalization.

3.4.7 Exchange Output Directory: Exchange_data

The `Exciton_data/Exchange_data/` directory contains run-level exchange datasets generated during exchange-enabled ***WannierBSE*** calculations. Created exclusively for workflows incorporating exchange interactions, this folder ensures strict computational traceability and data provenance for the current calculation. It aggregates the primary exchange tensors, retained lattice translation vectors, and auxiliary intermediate matrices evaluated during BSE Hamiltonian construction.

The primary files `R_mNN.mat` and `chi.mat` serve as run-level copies of the persistent cache files stored in `R_mNN_WBSE.mat` and `chi_WBSE.mat`, respectively. Because their internal array structures, dimensions, and index definitions are identical to those of the precomputed cache objects, their full formats are not duplicated here; refer to Section 3.3.3 for comprehensive definitions.

In addition to `R_mNN.mat` and `chi.mat`, this directory stores auxiliary run-level diagnostic quantities, including `G_all.mat` and `rho.mat`. Unlike the persistent cache in `Precomputed_data/Exchange_interaction/`, which is designed for reuse across multiple simulations, this directory acts as a self-contained output bundle directly tied to a specific BSE run, providing essential data for verification, post-processing diagnostics, and in-depth analysis of exchange-interaction matrix elements.

3.4.7.1 Reciprocal Vectors: G_all.mat

The MATLAB binary file `G_all.mat` stores the in-plane reciprocal-space vectors required during the exchange preprocessing stage. Generated automatically by ***WannierBSE***, this dataset is utilized to construct both the real-space overlap-density tensor `rho.mat` and the short-range exchange tensor `chi.mat`. Each retained vector G_m is formed from integer combinations of the primitive in-plane reciprocal lattice basis vectors b_1 and b_2:

$$G_m \;= n^{(G)}_{1,m} b_1 + n^{(G)}_{2,m} b_2$$

where $n^{(G)}_{1,m}$ and $n^{(G)}_{2,m}$ are integer indices labeling the m-th retained vector. The set of reciprocal vectors is governed by the user-defined cutoff parameter G_max, set in `exchange_control.txt`. During generation, candidate integer pairs are retained if their Cartesian magnitude satisfies $|\boldsymbol{G}| < G_{max}|b_1|$, with larger values of G_max incorporating higher-order reciprocal shell contributions.

The file contains a single real numeric matrix named G_all of dimension $\boldsymbol{N_G} \times \mathbf{2}$, where $\boldsymbol{N_G}$ represents the total number of retained nonzero reciprocal vectors. The first and second columns, G_all(:,1) and G_all(:,2), store the Cartesian $\boldsymbol{G_x}$ and $\boldsymbol{G_y}$ components, respectively, in standard reciprocal-space units (typically in Angstroms Å^{-1}). Notably, the uniform background component $\boldsymbol{G} = \mathbf{0}$ is explicitly excluded from this array, as the short-range exchange workflow acts strictly on the nonzero reciprocal components.

Example file:

```
%---           Gx           Gy
               0           -9.1172    % G1
               1.9739      -7.9775    % G2
               3.9478      -6.8379    % G3
               ...
               0            9.1172    % GNG
```

3.4.7.2 Pair-Density Tensor: rho.mat

The `rho.mat` file stores the intermediate Fourier-transformed Wannier pair-density tensor generated during the exchange preprocessing stage (see Eq. 2.10). ***WannierBSE*** evaluates this tensor by calculating the spinor overlaps between truncated Wannier functions and their real-space translated counterparts across the retained lattice translation vectors in `R_mNN.mat` and reciprocal lattice parameters in `G_all.mat`. This file serves as an intermediate exchange-preprocessing artifact consumed directly during the construction of `chi.mat`.

The file is a binary MATLAB data file containing a single complex double-precision tensor named rho. The rho array is five-dimensional, with array bounds structured as rho(N_{WF}, N_{WF}, N_z, N_R , N_G):

- Dimension 1: First Wannier basis function index $j' = 1, \cdots, N_{WF}$.
- Dimension 2: Second Wannier basis function index $j = 1, \cdots, N_{WF}$.
- Dimension 3: Truncated perpendicular z-grid slice z_i with $i = 1, \cdots, N_z$.
- Dimension 4: Retained real-space translation vectors $R = R_1, \cdots, R_{N_R}$.
- Dimension 5: Retained nonzero reciprocal vector $G = G_1, \cdots, G_{N_G}$.

Indices 1 and 2 correspond to the Wannier basis orbital pair forming the pair density. Dimensions 4 and 5 correspond strictly to the retained real-space translation vectors stored in `R_mNN.mat` and the nonzero reciprocal vectors in `G_all.mat`. Consequently, the translation dimension N_R and reciprocal dimension N_G in ϱ must match the row counts of `R_mNN.mat` and `G_all.mat` exactly.

4. Examples

This chapter illustrates the practical capabilities and operational modes of ***WannierBSE*** through hands-on tutorials located in the `Examples/` directory. Across all workflows, the solver supports flexible environmental dielectric screening, allowing users to choose between a simple, user-defined dielectric constant or the internal 2D Poisson solver to model complex dielectric environments in 2D systems.

To guide you through these operational modes, Chapter 5 organizes the tutorials around the interaction mechanisms included in the Bethe-Salpeter Equation:

- Direct Interaction Only: Focuses on baseline excitonic calculations driven solely by screened Coulomb attraction. In this mode, ***WannierBSE*** offers total electronic-structure flexibility, enabling users to seamlessly switch between full *ab initio* Hamiltonians derived from Wannier90 and custom, user-defined analytical tight-binding models.
- Direct + Short-Range Exchange: Covers the BSE workflow incorporating short-range electron-hole exchange interaction in addition to the direct interaction. Because evaluating exchange matrix elements requires explicit real-space Wannier function overlaps, this mode operates strictly on a full Wannier90 basis set.

4.1 Direct Coulomb Interaction Only

This section covers calculations where electron-hole pairs interact exclusively through the screened direct Coulomb potential. This operational mode is active when exchange interactions are disabled by setting *Exchange_Interaction* = *false* in `Parameters/exchange_control.txt` (see Section 3.1.3).

In this direct-only regime, ***WannierBSE*** provides maximum modeling versatility across both electronic structure inputs and dielectric screening approximations. For the underlying band structure, you may rely on full ab initio Hamiltonians derived from standard Wannier90 workflows or supply custom, user-defined analytical models ("Bring Your Own Tight-Binding Model"). For environmental screening, users can supply custom dielectric models, ranging from an effective dielectric constant to analytical dielectric functions, or evaluate screening directly through the internal 2D Poisson solver for complex 2D heterostructures.

The following four tutorials walk through these combinations, guiding you through parameter setup, file organization, and calculation execution for each operational mode.

4.1.1 Example 1: MoS_2 Heterostructure (hBN/MoS_2/SiO_2)

Directory: `01_hBN_MoS2_SiO2`

This example combines the *ab initio* accuracy of Wannier90 with the environmental screening simulated by the ***WannierBSE*** Poisson solver. We simulate a realistic experimental setup: a monolayer of MoS_2 supported by a Silicon Dioxide (SiO_2) substrate and covered by hexagonal Boron Nitride (hBN).

Key Input Files:

- `wannier90_hr.dat`: The tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- `wannier90.win`: The main Wannier90 input file, providing lattice vectors, atomic positions.

Parameter settings:

- `WTB_control.txt`: The top valence and the lowest conduction bands corresponding to the `wannier90_hr.dat` file are $VBM = 14$ and $CBM = 15$.
- `kmesh_control.txt`: The grid generation uses Nk_coarse=15 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=3 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=15, representing the number of k-points along a single direction in the dense sub-mesh.

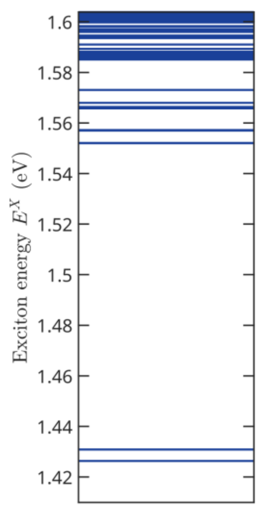
- `control.txt`: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- `WF_centers.txt`: Maps each of the 22 Wannier functions ($i = 1, \cdots, 22$) to an atomic center index $I \in \{1,2,3\}$ at position τ_I. For this MoS_2 monolayer, the three atomic positions in the unit cell correspond to Mo atom ($I = 1$), top sulfur atom ($I = 2$), and bottom sulfur atom ($I = 3$).
- `dielectric_control.txt`: To characterize the dielectric response of the MoS_2 monolayer, the intrinsic material properties are defined by a plasmon peak energy Epl=22.5 eV and in-plane dielectric constant ep_p=13.24. To model the encapsulation of the monolayer between hBN and the SiO_2, the heterostructure is configured with one capping layer and one substrate layer (t_layers=s_layers=1) characterized by dielectric constants of t_dielectric = 5.89 and s_dielectric= 3.9 respectively. The vertical geometry is established via layer_boundaries = 3.13, -3.13 (Å), which positions the hBN/MoS_2 and MoS2/SiO_2 interfaces maintaining consistency with the monolayer thickness d=6.26 Å specified in the `control.txt` file.

Exciton data:

Once the input files are prepared and the parameters are configured, the calculation is initiated by executing the main routine.
Upon running the WBSE.m main script, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- `Ex.mat`: This file contains the calculated exciton eigenvalues (in eV), ordered from lowest to highest. To visualize these results, use the auxiliary script `Ex_plot.m`, which automatically processes the .mat data to generate the energy spectrum plot shown below.

	1	
1	1.4263	E_1^X
2	1.4263	E_2^X
3	1.4308	E_3^X
4	1.4308	E_4^X
5	1.5520	E_5^X
6	...	



- `A.mat`: Stores the exciton wavefunctions (eigenvectors). Each column in this matrix represents the expansion coefficients of the excitonic state corresponding to the energies listed in `Ex.mat`.

	$A_1(vck)$	$A_2(vck)$	
	1	2	
1	1.0482e-10 + 1.1015e-10i	-6.8918e-11 + 7.1687e-11i	$v_1c_1k_1$
2	1.0082e-10 + 9.3671e-11i	-5.4049e-11 + 6.3244e-11i	$v_1c_1k_2$
3	1.3557e-10 + 4.4127e-11i	4.4762e-12 + 8.7065e-11i	$v_1c_1k_3$
4	1.8867e-10 + 3.2147e-11i	3.2750e-11 + 1.3221e-10i	$v_1c_1k_4$
5	1.3685e-10 + 1.2908e-10i	-7.5497e-11 + 1.1196e-10i	$v_1c_1k_5$
6	...	...	...

4.1.2 Example 2: MoS_2 in a complex Heterostructure (air/hBN/MoS_2/ice/SiO_2)

Directory: `02_air_hBN_MoS2_ice_SiO2`

In this example, we extend the Poisson solver's application to a multi-layered environment. This setup accounts for the hydrophilic nature of certain substrates, where an interfacial ice layer may form between the TMD and the support.

Specifically, we simulate a monolayer of MoS_2 on a Silicon Dioxide (SiO_2) with an intermediate 10 Å ice layer. The sample is capped with a hBN 10 nm layer and a semi-infinite air top-layer.

Key Input Files:

The user shall provide the primary input files:

- `wannier90_hr.dat`: In this case the user is only requested to provide the tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- `wannier90.win`: The main Wannier90 input file, providing lattice vectors, atomic positions.

Parameter settings:

- `WTB_control.txt`: The top valence band and to lowest conduction bands, corresponding to the `wannier90_hr.dat` file, are $VBM = 14$ and $CBM = 15$.
- `structure.txt`: Contains the lattice vectors and atomic positions of the MoS_2 unit cell, matching the geometry used to generate `wannier90_hr.dat`.
- `kmesh_control.txt`: The grid generation uses Nk_coarse=15 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=3 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=15, representing the number of k-points along a single direction in the dense sub-mesh.

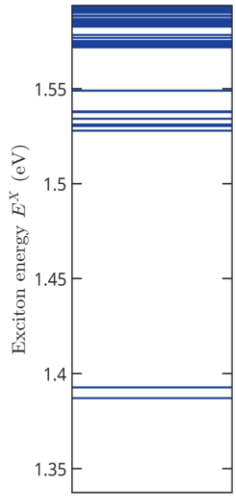
- `control.txt`: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- `WF_centers.txt`: Maps each of the 22 Wannier functions ($i = 1, \cdots, 22$) to an atomic center index $I \in \{1,2,3\}$ at position τ_I. For this MoS_2 monolayer, the three atomic positions in the unit cell correspond to Mo atom ($I = 1$), top sulfur atom ($I = 2$), and bottom sulfur atom ($I = 3$).
- `dielectric_control.txt`: The dielectric response of the MoS_2 monolayer is defined by its intrinsic material properties, using a plasmon peak energy Epl=22.5 eV and in-plane dielectric constant of ep_p=13.24. To represent the complex environmental screening of this heterostructure, the system is configured with two capping layers (air and hBN) and two substrate layers (ice and SiO_2), set via t_layers=s_layers=2, respectively. Ordered from the top-most layer to the bottom-most, the dielectric constants are defined as t_dielectric = 1, 5.89 and s_dielectric=1.9, 3.9. The vertical geometry is established via layer_boundaries = 103.13, 3.13, -3.13, -13.13 (Å). These coordinates define the interfaces for air/hBN, hBN/MoS_2, MoS2/ice, and ice/SiO_2. This configuration ensures consistency with a 10 nm hBN capping layer, a 10 Å interfacial ice layer, and an effective monolayer thickness of d=6.26 Å, which is specified in the `control.txt` file.

Exciton data:

Once the input files are prepared and the parameters are configured, the calculation is initiated by executing the main routine.
Upon running the `WBSE.m` main script, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- `Ex.mat`: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.3871	E_1^X
2	1.3871	E_2^X
3	1.3927	E_3^X
4	1.3927	E_4^X
5	1.5278	E_5^X
6	...	



- `A.mat`: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vck)$	$A_2(vck)$	
	1	2	
1	-9.5057e-12 + 1.309e-11i	3.0257e-12 + 8.407e-11i	$v_1c_1k_1$
2	4.8988e-12 + 1.128e-11i	8.5658e-12 + 6.752e-11i	$v_1c_1k_2$
3	3.9398e-11 + 4.797e-11i	6.8644e-11 + 4.859e-11i	$v_1c_1k_3$
4	5.3161e-11 + 9.056e-11i	1.2008e-10 + 6.164e-11i	$v_1c_1k_4$
5	-2.8049e-11 + 4.783e-11i	2.5553e-11 + 1.200e-10i	$v_1c_1k_5$
6	...	...	...

4.1.3 Example 3: MoS_2 using an Analytic Tight-Binding Model

Directory: `03_MoS2_analytic_TB_model`

This example demonstrates ***WannierBSE***'s ability to interface directly with external or custom analytical models, bypassing Wannier90 entirely. Note that this custom-model workflow is available exclusively when evaluating direct Coulomb interactions only. As example, we calculate the excitonic spectrum of Monolayer MoS2 using the two-band tight-binding model described by Berkelbach *et al.* [22].

Here, we utilize the internal Poisson solver to compute the dielectric screening, assuming the monolayer is suspended in a vacuum (Air/MoS_2/Air). This example highlights how users can provide their own Hamiltonian matrix elements and k-point meshes directly.

Key Input Files:

The user shall provide the primary input files:

- `c1_TB.txt` & `v1_TB.txt`: Energy dispersion files for the conduction (`c1_TB.txt`) and valence bands (`v1_TB.txt`), respectively. These files must be located inside the `User_input/TB_data/` directory.
- `kmesh.txt`: Reciprocal-space coordinate mesh defining the $\boldsymbol{k}$-point grid that corresponds to the energy dispersion data.
- `structure.txt`: Lattice vectors and atomic positions for the monolayer MoS_2 unit cell, matching the geometry used to generate the band structure.

Parameter settings:

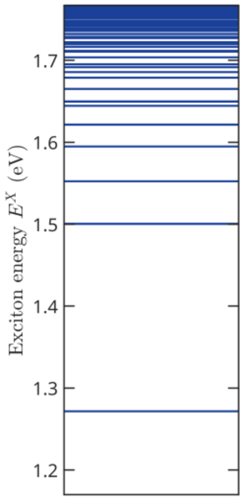
- `control.txt:` The MoS_2-ML thickness is set $d = 6.26\,(\text{Å})$. The BSE Hamiltonian is constructed using 1 valence and 1 conduction bands (Nv=1, Nc=1). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- `WF_centers.txt`: Following the 2-band model described by Berkelbach et al., the Hamiltonian basis is constructed from two orbitals localized on the transition-metal atom $(I = 1)$.These include the $|d_{z^2}\rangle \equiv |\phi_c\rangle$ orbital for the conduction band and the $|d_{x^2-y^2}\rangle + i\tau|d_{xy}\rangle \equiv |\phi_v^\tau\rangle$ orbital for the valence band.
- `dielectric_control.txt`: To calculate the dielectric function of a suspended MoS_2 monolayer, the material is characterized by a plasmon peak energy Epl = 22.5 eV and an in-plane dielectric constant ep_p = 13.24. To simulate the vacuum or air environment surrounding the monolayer, the system is configured with a single capping and substrate layer (t_layers = 1, s_layers = 1) both assigned a dielectric constant of t_dielectric = s_dielectric = 1.0. In accordance with a monolayer thickness of d = 6.26 Å, the air/MoS_2 and MoS_2/air interfaces are symmetrically positioned via layer_boundaries = 3.13, -3.13 (Å).

Exciton data:

Similarly to the previous example, upon running the main script `WBSE.m` main, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- `Ex.mat:` This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.2716	E_1^X
2	1.5003	E_2^X
3	1.5005	E_3^X
4	1.5524	E_4^X
5	1.5948	E_5^X
6	...	



- A.mat: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vck)$	$A_2(vck)$	
	1	2	
1	0.0062 - 0.0106i	-1.2060e-05 + 5.5577e-06i	$v_1c_1k_1$
2	0.0060 - 0.0107i	1.19791e-05 - 5.7302e-06i	$v_1c_1k_2$
3	0.0066 - 0.0107i	-5.1541e-05 + 2.2425e-05i	$v_1c_1k_3$
4	0.0063 - 0.0107i	-2.5448e-05 + 1.1506e-05i	$v_1c_1k_4$
5	0.0060 - 0.0108i	2.5107e-05 - 1.2232e-05i	$v_1c_1k_5$
6	...	...	

4.1.4 Example 4: MoS_2 with Constant Dielectric Function

Directory: `04_MoS2_constant_epsilon`

This example demonstrates how ***WannierBSE*** allows users to integrate their own custom dielectric models. Here, we apply this flexibility to study a freestanding MoS_2 monolayer by adopting a constant effective dielectric screening, implemented within the framework of the local screening approximation.

In this configuration, the internal Poisson solver is disabled. Instead, the user manually defines the screening environment via the `epsilon.txt` file.

Key Input Files:

The calculation relies on the following primary input files:

- `wannier90_hr.dat`: The tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- `wannier90.win`: The main Wannier90 input file, providing lattice vectors, atomic positions.
- `epsilon.txt`: A user-provided file defining the dielectric function based on his own model used to screen the Coulomb interaction, in this case a constant effective dielectric constant ϵ_{eff}=7.7.

Parameter settings:

A summary of the parameter's configuration set in for this example is provided below:

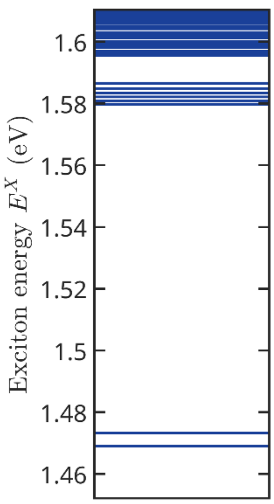
- `WTB_control.txt`: The top valence band and to lowest conduction band corresponding to the `wannier90_hr.dat` are VBM=14 and CBM =15.
- `kmesh_control.txt`: The grid generation uses Nk_coarse=15 to set the base resolution along the $\Gamma - K$ line. The high-density coverage is defined by Dense_range=3 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=15, representing the number of k-points along a single direction in the dense sub-mesh.
- `WF_centers.txt`: The corresponding index I of each atomic center I where the i-th Wannier function is localized, used during the Wannier90 calculations is in the `WF_centers.txt` file.
- `control.txt`: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).

Exciton data:

Following the same procedure as the previous example, executing the main script WBSE.m initiates the construction and diagonalization of the BSE Hamiltonian. The calculated excitonic properties are subsequently stored in the following output files:

- `Ex.mat`: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.46909	E_1^X
2	1.46909	E_2^X
3	1.47325	E_3^X
4	1.47325	E_4^X
5	1.57966	E_5^X
6	...	



- `A.mat`: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vck)$		$A_2(vck)$		
	1		2		
1	-5.771e-11	-6.530e-11i	3.289e-11	+1.546e-10i	$v_1c_1k_1$
2	-6.320e-11	-4.974e-11i	3.713e-11	+1.322e-10i	$v_1c_1k_2$
3	-1.013e-10	-2.737e-11i	1.068e-10	+1.0091e-10i	$v_1c_1k_3$
4	-1.474e-10	-2.801e-11i	1.7094e-10	+1.169e-10i	$v_1c_1k_4$
5	-7.903e-11	-8.8516e-11i	5.9698e-11	+1.933e-10i	$v_1c_1k_5$
6	…. . .		. . . .		

4.2 Direct and Short-Range Exchange Coulomb Interactions

This section covers calculations that incorporate both direct screened Coulomb attraction and short-range electron-hole exchange interactions. This operational regime is enabled by setting Exchange_Interaction = true in `Parameters/exchange_control.txt` (see Section 3.1.3).

Because evaluating exchange matrix elements requires explicit spatial overlaps between localized Wannier orbitals, exchange-enabled calculations strictly require Wannier90-based inputs rather than user-provided external tight-binding models. To maximize computational efficiency, ***WannierBSE*** leverages persistent data caching: it reuses preprocessed Wannier function spatial data when available and, where applicable, reloads precomputed exchange tensors by setting `Use_Precomputed_Chi = true`. In this mode, ***WannierBSE*** expects `chi_WBSE.mat` and `R_mNN_WBSE.mat` to be present in `Precomputed_data/Exchange_interaction/`.

The following three tutorials demonstrate these exchange-enabled capabilities across different dielectric screening environments and data-reuse workflows.

4.2.1 Example 5: hBN/MoS_2/SiO_2 considering short-range exchange interaction

Directory: 05_hBN_MoS2_SiO2_Dir_Short_Exch

In this example, we extend the baseline setup of Example 5.1.1 by incorporating short-range exchange corrections alongside direct Coulomb screening from the internal 2D Poisson solver. This tutorial walks through a complete, end-to-end calculation performed entirely from scratch, serving as the baseline reference before introducing data-reuse optimizations.

Key Input Files:

- `wannier90_hr.dat`: The tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- `wannier90.win`: The main Wannier90 input file, providing lattice vectors, atomic positions.
- `Wannier_functions_xsf`: Directory containing the Wannier functions in XSF format, organized into four component subfolders: up_real/, up_imag/, down_real/, and down_imag/.

Parameter settings:

- `WTB_control.txt`: The top valence and the lowest conduction bands corresponding to the wannier90_hr.dat file are VBM=14 and CBM =15.
- `kmesh_control.txt`: The grid generation uses Nk_coarse=15 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=3 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=15, representing the number of k-points along a single direction in the dense sub-mesh.
- `control.txt`: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- `WF_centers.txt`: Maps each of the 22 Wannier functions (i=1,,22) to an atomic center index I={1,2,3} at positionτ_I. For this MoS_2 monolayer, the three atomic positions in the unit cell correspond to Mo atom (I=1), top sulfur atom (I=2), and bottom sulfur atom (I=3).
- `dielectric_control.txt`: To characterize the dielectric response of the MoS_2 monolayer, the intrinsic material properties are defined by a plasmon peak energy Epl=22.5 eV and in-plane dielectric constant ep_p=13.24. To model the encapsulation of the monolayer between hBN and the SiO_2, the heterostructure is configured with one capping layer and one substrate layer (t_layers=s_layers=1) characterized by dielectric constants of t_dielectric = 5.89 and s_dielectric= 3.9 respectively. The vertical geometry is established via layer_boundaries = 3.13, -3.13 (Å), which positions the hBN/MoS_2 and MoS2/SiO_2 interfaces maintaining consistency with the monolayer thickness d=6.26 Å specified in the control.txt file.
- `Exchange_control.txt`: The short-range electron-hole exchange interaction is enabled (`Exchange_Interaction=true`). The `chi_WBSE.mat` and `R_mNN_WBSE.mat` exchange quantities are calculated from the Wannier-function data (`Use_Precomputed_Chi=false`), and the raw XSF files are retained after processing for inspection or visualization (`cleanup_xsf=false`).

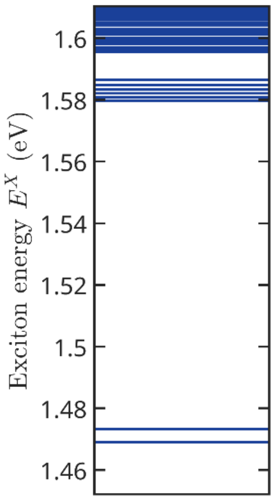
Exciton data:

Once the input files are prepared and the parameters are configured, the calculation is initiated by executing the main routine.

Upon running the `WBSE.m` main script, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- Ex.mat: This file contains the calculated exciton eigenvalues (in eV), ordered from lowest to highest. To visualize these results, use the auxiliary script `Ex_plot.m`, which automatically processes the .mat data to generate the energy spectrum plot shown below.

	1	
1	1.42637	E_1^X
2	1.42638	E_2^X
3	1.44108	E_3^X
4	1.44109	E_4^X
5	1.55200	E_5^X
6	...	



- `A.mat`: Stores the exciton wavefunctions (eigenvectors). Each column in this matrix represents the expansion coefficients of the excitonic state corresponding to the energies listed in `Ex.mat`.

	$A_1(vck)$	$A_2(vck)$	
	1	2	
1	-1.1312e-09 - 3.4039e-09i	-1.8837e-09 + 2.4969e-09i	$v_1c_1k_1$
2	-1.0822e-09 - 3.4142e-09i	-1.9212e-09 + 2.4393e-09i	$v_1c_1k_2$
3	-9.6717e-10 - 3.0150e-09i	-1.7921e-09 + 2.2192e-09i	$v_1c_1k_3$
4	-1.0059e-09 - 3.1298e-09i	-1.8800e-09 + 2.2369e-09i	$v_1c_1k_4$
5	-1.0215e-09 - 3.2520e-09i	-1.8428e-09 + 2.2369e-09i	$v_1c_1k_5$
6	...	...	...

4.2.2 Example 6: air/hBN/MoS_2/ice/SiO_2 considering short-range exchange interaction

Directory: 06_air_hBN_MoS2_ice_SiO2_Dir_Short_Exch

Here, we extend the multi-layer setup introduced in Example 5.1.2 to incorporate short-range exchange interactions. This example highlights workflow optimization in ***WannierBSE***, showing how to bypass intensive intermediate evaluations by loading precomputed screening matrices (`chi_WBSE.mat`) and preprocessed spatial Wannier orbitals from `Precomputed_data/` directory.

Key Input Files:

The user shall provide the primary input files:

- `wannier90_hr.dat`: In this case the user is only requested to provide the tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- `Wannier_functions_xsf`: Directory containing the Wannier functions in XSF format, organized into four component subfolders: up_real/, up_imag/, down_real/, and down_imag/.
- `wannier90.win`: The main Wannier90 input file, providing lattice vectors, atomic positions.

Parameter settings:

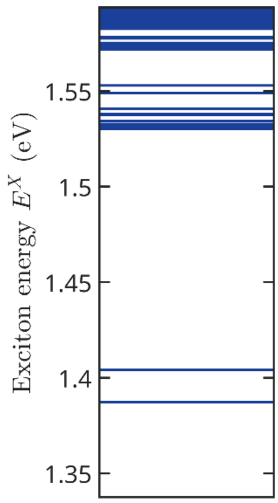
- `WTB_control.txt`: The top valence band and to lowest conduction bands, corresponding to the `wannier90_hr.dat` file, are VBM=14 and CBM =15.
- `structure.txt`: Contains the lattice vectors and atomic positions of the MoS_2 unit cell, matching the geometry used to generate `wannier90_hr.dat`.
- `kmesh_control.txt`: The grid generation uses Nk_coarse=15 to set the base resolution along the $\Gamma - K$ line. The high-density coverage is defined by Dense_range=3 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=15, representing the number of k-points along a single direction in the dense sub-mesh.
- `control.txt`: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- `WF_centers.txt`: Maps each of the 22 Wannier functions ($i = 1, \dots, 22$) to an atomic center index I= {1,2,3} at position τ_I. For this MoS_2 monolayer, the three atomic positions in the unit cell correspond to Mo atom (I=1), top sulfur atom (I=2), and bottom sulfur atom (I=3).
- `dielectric_control.txt`: The dielectric response of the MoS_2 monolayer is defined by its intrinsic material properties, using a plasmon peak energy Epl=22.5 eV and in-plane dielectric constant of ep_p=13.24. To represent the complex environmental screening of this heterostructure, the system is configured with two capping layers (air and hBN) and two substrate layers (ice and SiO_2), set via t_layers=s_layers=2, respectively. Ordered from the top-most layer to the bottom-most, the dielectric constants are defined as t_dielectric = 1, 5.89 and s_dielectric=1.9, 3.9. The vertical geometry is established via layer_boundaries = 103.13, 3.13, -3.13, -13.13 (Å). These coordinates define the interfaces for air/hBN, hBN/MoS_2, MoS2/ice, and ice/SiO_2. This configuration ensures consistency with a 10 nm hBN capping layer, a 10 Å interfacial ice layer, and an effective monolayer thickness of d=6.26 Å, which is specified in the `control.txt` file.
- `exchange_control.txt`: The short-range electron-hole exchange interaction is enabled (Exchange_Interaction=true). Precomputed exchange quantities are loaded by setting Use_Precomputed_Chi=true, avoiding the recalculation of the chi and RmNN tensors. The temporary/raw XSF files are removed after processing (cleanup_xsf=true).

Exciton data:

Once the input files are prepared and the parameters are configured, the calculation is initiated by executing the main routine. Upon running the `WBSE.m` main script, the code constructs the BSE Hamiltonian and performs the diagonalization. The resulting excitonic properties are saved to the following output files:

- `Ex.mat`: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.38715	E_1^X
2	1.38717	E_2^X
3	1.40405	E_3^X
4	1.40406	E_4^X
5	1.53034	E_5^X
6	...	



- `A.mat`: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k).

	$A_1(vck)$	$A_2(vck)$	
	1	2	
1	5.0227e-10 + 3.1821e-09i	2.0273e-09 + 1.9001e-09i	$v_1c_1k_1$
2	4.5587e-10 + 3.1833e-09i	1.9713e-09 + 1.9291e-09i	$v_1c_1k_2$
3	4.1888e-10 + 2.8477e-09i	1.8131e-09 + 1.8201e-09i	$v_1c_1k_3$
4	4.3434e-10 + 2.9339e-09i	1.8054e-09 + 1.8850e-09i	$v_1c_1k_4$
5	4.3270e-10 + 3.0531e-09i	1.8115e-09 + 1.8534e-09i	$v_1c_1k_5$
6	...	...	...

4.2.3 Example 7: MoS_2 with Constant Dielectric Function considering short-range exchange interaction

Directory: 07_MoS2_constant_epsilon_Dir_Short_Exch

Finally, this tutorial applies the exchange-enabled workflow to the uniform screening regime introduced in Example 5.1.4. Following the caching methodology from the previous example, it showcases how reloading precalculated `chi_WBSE.mat` tensors and preprocessed Wannier function spatial data enables rapid parameter sweeps under an effective dielectric constant.

Key Input Files:

The calculation relies on the following primary input files:

- `wannier90_hr.dat`: The tight-binding Hamiltonian generated by a standard Wannier90 calculation.
- `wannier90.win`: The main Wannier90 input file, providing lattice vectors, atomic positions.
- `Wannier_functions_xsf`: Directory containing the Wannier functions in XSF format, organized into four component subfolders: up_real/, up_imag/, down_real/, and down_imag/.
- `epsilon.txt`: A user-provided file defining the dielectric function based on his own model used to screen the Coulomb interaction, in this case a constant effective dielectric constant eff=7.7.

Parameter settings:

A summary of the parameter's configuration set in for this example is provided below:

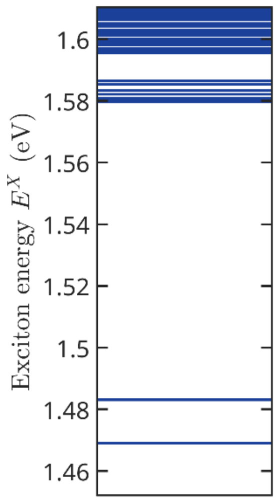
- `WTB_control.txt`: The top valence band and to lowest conduction band corresponding to the `wannier90_hr.dat` are VBM=14 and CBM =15.
- `structure.txt`: Contains the lattice vectors and atomic positions of the MoS_2 unit cell, matching the geometry used to generate `wannier90_hr.dat`.
- `kmesh_control.txt`: The grid generation uses Nk_coarse=15 to set the base resolution along the Γ to K line. The high-density coverage is defined by Dense_range=3 (in coarse-grid units), while the sampling density within those regions is set by Nk_dense=15, representing the number of k-points along a single direction in the dense sub-mesh.
- `WF_centers.txt`: The corresponding index I of each atomic center I where the i-th Wannier function is localized, used during the Wannier90 calculations is in the `WF_centers.txt file`.
- `control.txt`: The BSE Hamiltonian is constructed using 2 valence and 2 conduction bands (Nv=2, Nc=2). The solver is configured to compute the lowest 100 excitonic eigenvalues (NVAL=100).
- `exchange_control.txt`: The short-range electron-hole exchange interaction is enabled (Exchange_Interaction=true). Precomputed exchange quantities are loaded by setting Use_Precomputed_Chi=true, avoiding the recalculation of the chi and RmNN tensors.

Exciton data:

Following the same procedure as the previous example, executing the main script WBSE.m initiates the construction and diagonalization of the BSE Hamiltonian. The calculated excitonic properties are subsequently stored in the following output files:

- Ex.mat: This file lists the calculated exciton energies (in eV), ordered from lowest to highest.

	1	
1	1.46911	E_1^X
2	1.46912	E_2^X
3	1.48309	E_3^X
4	1.48309	E_4^X
5	1.57966	E_5^X
6	...	



- `A.mat`: As described in the previous section, this file stores the exciton eigenvectors. The matrix represents the expansion coefficients of the exciton states within the transition basis (v, c, k)

	A1vck	A2vck	
	1	2	
1	-2.2383e-09 - 3.1501e-09i	3.3307e-09 + 2.7838e-10i	$v_1c_1k_1$
2	-2.1967e-09 - 3.1824e-09i	3.3025e-09 + 3.4696e-10i	$v_1c_1k_2$
3	-1.9296e-09 - 2.7716e-09i	2.9901e-09 + 3.5150e-10i	$v_1c_1k_3$
4	-2.0227e-09 - 2.8988e-09i	3.0704e-09 + 4.1592e-10i	$v_1c_1k_4$
5	-2.0680e-09 - 3.0150e-09i	3.0551e-09 + 3.9026e-10i	$v_1c_1k_5$
6	...	...	...

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